

# MAPLE: Metadata-Aware Pretraining for Localized Knowledge Disambiguation

Anonymous ACL submission

## Abstract

Many questions cannot be answered from text alone because the correct answer is defined by local institutions, conventions, or recent events. We study this problem as *localized knowledge disambiguation*, where an under-specified question can require different answers under different locales. We introduce LOCALNEWSQA, a reusable English-news benchmark with row-level source validation that holds question frames fixed across paired target and contrast locales, separating localized factual recall from answer switching under locale changes. Because target-locale accuracy can miss models that give the same answer after a locale change, paired switching is the primary metric. We also introduce MAPLE, a controlled family of decoder-only models pretrained with corpus-native geographic provenance, and compare it to metadata-free controls trained with the same data, token budgets, architectures, and optimization. Across 33 models trained from scratch at 0.5B and 1B on the NOW news corpus (Davies, 2016), with global comparisons extended to 3B, metadata-free controls record 0/16,409 paired locale-switch successes, while MAPLE produces measurable exact-pair and margin-switch behavior and improves ambiguous-item accuracy by up to +7.9 points. Together, perplexity, ablation, coverage, and external-benchmark analyses support a modest pretraining-dominant interpretation in which corpus-native provenance helps one global model make more locale-sensitive predictions; token-efficiency curves show the benefit persists from 500M to 3B.<sup>1</sup>

## 1 Introduction

Large language models are trained on text from many countries, institutions, and communities, yet they are usually optimized as if this evidence belonged to one broad distribution. This makes them

<sup>1</sup>Code is available at [https://anonymous.4open.science/r/metadata\\_localization](https://anonymous.4open.science/r/metadata_localization)

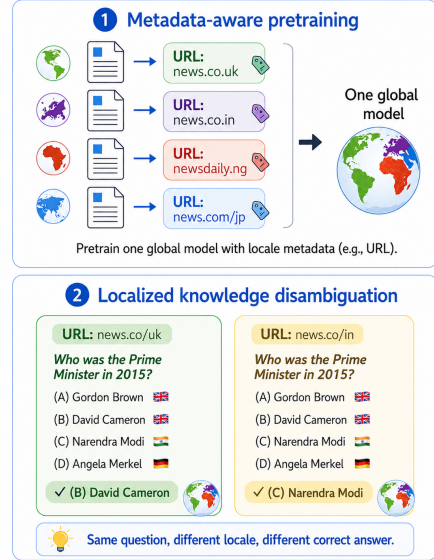


Figure 1: MAPLE uses web-native provenance metadata during pretraining so that one global model can condition on locale metadata at evaluation time.

fluent generalizers, but it can also produce geographically homogenized behavior and “western-aligned” defaults (Weidinger et al., 2022; Ghosh et al., 2025). The resulting failure is not only lack of local knowledge. A model may encode facts from many places and still fail to bind a fact to the locale where it is valid.

That binding failure becomes visible in under-specified questions. A question about the department that reports census statistics, the school years covered by “primary school”, or the sport implied by “football” may have a locally obvious answer, but the correct answer changes across countries. Human readers resolve such questions from context or ask for clarification. A model trained on global text should similarly use an available locale signal to choose the locally valid answer. We call this problem *localized knowledge disambiguation*. It draws on under-specified question answering (Parrish et al., 2022), geographic QA (Yin et al., 2022; Hofmann et al., 2024), cultural evaluation

(Adilazuarda et al., 2024; Choenni et al., 2024), and geography-aware adaptation (Cao et al., 2025), but it isolates a sharper behavior. The model must choose among locale-specific answer distributions rather than produce one global default.

Existing evaluations make important progress on geographically and culturally grounded behavior, but they usually score isolated facts, opinions, values, norms, or culturally situated tasks. For localized disambiguation, target-locale accuracy is incomplete because a model can answer one locale correctly while giving the same answer when the locale changes. LOCALNEWSQA differs from GEOMLAMA-style geographic probes and culture or opinion benchmarks by holding the question frame fixed across paired target and contrast locales, making answer switching part of the evaluation rather than a post-hoc analysis. Existing localization methods also tend to intervene after pretraining through prompting, fine-tuning, continued pretraining, or geography-aware objectives (Gururangan et al., 2020; Adilazuarda et al., 2024; Cao et al., 2025; Hofmann et al., 2024). Those methods are useful, but they leave open a more basic question. If training documents already contain usable provenance, can a model learn during pretraining to preserve multiple local answer distributions in a form that can later be accessed?

Although we study this failure mode through geographic provenance in English news, the underlying issue is broader. Whenever training data mixes multiple context-conditioned answer distributions, a model may know several plausible answers yet fail to bind the correct one to the active context. Geography gives us an observable and externally verifiable testbed for this behavior, so we use it here as a clean instance of a broader context-binding problem.

We build the study around two artifacts. LOCALNEWSQA is a reusable English-news benchmark with explicit localized-recall items, ambiguous locale-dependent items, and paired target/contrast examples whose gold answers differ. This design separates ordinary localized factual recall from the stronger requirement that a model switch answers when the locale changes. MAPLE, Metadata Aware Pretraining for Localized Evaluation, is a controlled family of decoder-only models trained from scratch with corpus-native provenance metadata from the NOW corpus (Davies, 2016). The intervention is deliberately simple. It prepends tex-

tual metadata to documents and asks whether that signal changes what one global model can retain and access under fixed data, architecture, optimizer, and token-budget choices.

The contribution is not a new kind of metadata tag, nor a claim that LOCALNEWSQA replaces broad geographic or cultural evaluation. Instead, we isolate localized knowledge disambiguation as an important failure mode, use paired switching as its central metric, and test whether corpus-native provenance helps one global model access local knowledge. The experiments then ask whether this behavior appears in paired QA, held-out language modeling, provenance and coverage ablations, scaling runs, and adjacent benchmarks.

The paper makes four contributions.

- **Task and benchmark.** We define localized knowledge disambiguation as a missing evaluation target and introduce LOCALNEWSQA, a reusable 35,874-item English-news benchmark with explicit and ambiguous splits, paired target/contrast examples, row-level source-validation fields, and a stratified 200-item human audit.
- **Controlled pretraining.** We introduce MAPLE, a controlled study using corpus-native URL, country, and continent provenance while holding data, tokenizer, architecture, optimizer, and token budget fixed.
- **Paired switching results.** Across 33 from-scratch 0.5B/1B models plus 3B scaling runs, MAPLE improves most on ambiguous items and produces paired answer switching where the metadata-free control records no successful switches.
- **Mechanism and scope.** We explain the effect through local/global perplexity, metadata ablations, omitted-region controls, train/inference factorial comparisons, six adjacent external benchmarks, and 500M-to-3B token-efficiency gains.

## 2 Related Work

Localized knowledge disambiguation intersects under-specified question answering, locale-sensitive prompting, culturally grounded evaluation, and geographic QA. Prior work shows that models encode uneven cultural and geographic assumptions and that prompting or adaptation can induce more locale-sensitive behavior (Parrish et al., 2022; Yin et al., 2022; Adilazuarda et al.,

2024; Cao et al., 2025; Choenni et al., 2024; Hofmann et al., 2024). LOCALNEWSQA isolates a different object of measurement. It asks whether the same question frame resolves to different answers under different locales.

Metadata conditioning has been used to steer or contextualize models with control codes, time, language, document, source, and social-context signals (Keskar et al., 2019; Dhingra et al., 2022; Liu et al., 2020; Aghajanyan et al., 2022; Korbak et al., 2023; Khalifa et al., 2024; Weller et al., 2024; Kulkarni et al., 2021; Li et al., 2022). CTRL (Keskar et al., 2019) studies steerable generation through a fixed curated code inventory. LMSOC (Kulkarni et al., 2021) augments encoder-side representations with learned social-context embeddings. NTULM (Li et al., 2022) introduces non-textual user signals for encoder pretraining on social media. MAPLE asks whether corpus-native geographic provenance learned during decoder-only pretraining becomes an inference-time control variable for locale-conditioned answer selection under paired metadata and no-metadata controls.

Our work is closest to Gao et al. (2025), who show that metadata conditioning with real URLs can accelerate modern decoder-only pretraining. We build on that insight and ask a different question. Their main outcome is general language-modeling quality. Our outcome is whether geographic provenance helps one model preserve and access multiple locale-specific answer distributions. A method can improve average pretraining loss without improving localized disambiguation, and a locale cue can help at inference without having been learned as a usable pretraining signal. MAPLE studies both sides through train-time and inference-time metadata controls.

### 3 Task and Benchmark

We first formalize the behavior that paired locale switching is meant to capture, then show how LOCALNEWSQA turns it into a reusable benchmark.

#### 3.1 What Localized Knowledge Disambiguation Requires

Localized knowledge disambiguation evaluates whether a model can condition a factual answer on the locale where the fact applies. The key difficulty is that the question need not name the locale. If a prompt asks which agency releases census numbers, which school years count as primary school,

or which league is implied by a sports term, the globally most common answer can be wrong for the intended locale. The diagnostic examples are therefore not merely country-specific questions, but questions whose answer changes when the locale changes.

LOCALNEWSQA is designed around that requirement. The *explicit* split names the locale in the question and measures localized factual recall. The *ambiguous* split omits the locale from the question text and supplies the locale through the evaluation context. Each ambiguous row also has a paired contrast locale with a different gold answer. This pairing makes LOCALNEWSQA a direct testbed for the behavior that accuracy-only evaluations miss, namely whether the same model changes its answer when the locale changes.

#### 3.2 LOCALNEWSQA Construction and Validation

We construct LOCALNEWSQA in English news because news provides dense local institutions, public bodies, sports, education systems, media references, and recent events with verifiable provenance. We generate candidates through stratified GPT-5.4 overgeneration across country, split, topic, and year shards. The retained candidate set contains 35,874 questions from 17 countries across 4 continents, split into *explicit* (19,465) and *ambiguous* (16,409) items over 8 topic families, 🏛️ politics/government, 🏢 civic institutions, 📈 economy/business, 🎓 education, 🎭 media/culture, ⚽ sports, 🎉 public life/holidays, and 🗺️ geography/infrastructure.

The construction pipeline deliberately overgenerates before pruning. It begins with 41,002 candidates and removes 0.7% by exact and near deduplication, followed by an 11.9% structured-pruning pass. The pruning stage enforces option integrity, answer consistency, ambiguity validity, low-information filtering, and low-leakage formatting. For ambiguous items, rows that mention the target country in the question text are removed, because they no longer test disambiguation.

Every retained row is passed through split-specific automatic source validation. Explicit rows require target-side certification, while ambiguous rows require both target and contrast certification. This yields a 35,870-row fully certified subset within the 35,874-row retained benchmark; the four unresolved ambiguous rows are flagged in the release. We also conduct a stratified 200-item

human audit over four countries, one per continent. Annotators see the question, answer options, locale tags, evidence links, and evidence snippets, and they judge target correctness, contrast correctness, and whether the answer genuinely changes across locales. Appendix A.3 gives the full construction, validation, and human-audit protocol.

### 3.3 LOCALNEWSQA Evaluation

We report target-locale accuracy for comparability, but for this task it is not sufficient. The defining requirement of localized knowledge disambiguation is that the same question frame can resolve differently under different locales, so paired switching is the primary metric and target accuracy is a one-sided view. For an ambiguous pair with question frame  $q$ , target locale  $\ell_t$ , contrast locale  $\ell_c$ , and locale-specific answers  $a_t, a_c$ , target accuracy only asks whether the model chooses  $a_t$  under  $\ell_t$ . *Exact pair* requires the same model to choose  $a_t$  under  $\ell_t$  and  $a_c$  under  $\ell_c$ . *Margin switch* requires  $s(a_t | q, \ell_t) > s(a_c | q, \ell_t)$  and  $s(a_c | q, \ell_c) > s(a_t | q, \ell_c)$ . Exact pair is intentionally stringent because both sides of the same question frame must be answered correctly. Margin switch is a softer but still paired test of whether the model’s score ordering moves toward the locale-appropriate answer on both sides.

For MAPLE, we score pretrained checkpoints before instruction tuning using closed-form option scoring. For the main LOCALNEWSQA comparison, we fix the answer-order seed to 41 for all models and tables, and use question-level bootstrap intervals; the seed is not chosen per model or per result. We score each candidate option text under the prompt and predict by argmax over option scores, isolating what pretraining encodes before task-specific instruction tuning. Appendix Figure 12 gives the exact prompts, and the appendix repeats the comparison after LoRA instruction tuning. LOCALNEWSQA is meant to be reusable beyond MAPLE: its splits separate localized recall from paired switching, its target/contrast structure is model-agnostic, and its release includes prompts and row-level source-validation fields.

## 4 MAPLE Controlled Pretraining

We use the NOW corpus (Davies, 2016) throughout. Each document includes a publication year, source URL, and country of origin, which we map to continents and serialize as structured metadata before

| # Continents | Train Meta | Eval Meta | Notation      |
|--------------|------------|-----------|---------------|
| 1            | ✓          | ✓         | LOCAL(T+,I+)  |
| 1            | ✓          | ✗         | LOCAL(T+,I-)  |
| 1            | ✗          | ✓         | LOCAL(T-,I+)  |
| 1            | ✗          | ✗         | LOCAL(T-,I-)  |
| 4            | ✓          | ✓         | GLOBAL(T+,I+) |
| 4            | ✗          | ✗         | GLOBAL(T-,I-) |

Table 1: Model variants used in this paper. “T+/T-” indicate metadata present/absent during training; similarly “I+/I-” indicate metadata available versus not available at inference. (Bordered boxes) also indicate I+ models.

the article title and content. After filtering, the corpus contains approximately 4M documents from Africa, 10M from America, 7.5M from Asia, and 5.6M from Europe. All local and global models are trained on 41.9B tokens. Local models sample one continent to that budget, while global models mix all four continents before sampling. Appendix A.1 gives the full data formatting and geographic coverage details.

MAPLE uses the simplest intervention that can answer our pretraining question. Metadata-formatted documents prepend source URL, country, and continent fields before the title and article text. Metadata-free documents omit these fields. The comparison is deliberately controlled so that changes in behavior can be attributed to the presence of provenance metadata rather than to data volume, architecture, optimizer, or tokenizer changes.

The 33 from-scratch 0.5B/1B models include 10 core 500M models (4 continents  $\times$  2 train-metadata settings plus 2 global settings) and 23 1B models with the same 10 core settings, 5 provenance-granularity ablations, and 8 leave-one-continent controls. The 3B results add two global scaling runs.

Throughout the results, T+/T- indicate metadata present or absent during training, and I+/I- indicate metadata present or absent at inference. The notation is factorial. It separates whether a model has learned from metadata during pretraining from whether metadata is supplied in the evaluation context. We introduce the notation here and use it sparingly in the text when the comparison would otherwise be ambiguous.

## 5 Results

We organize the evidence from the target behavior outward, moving from LOCALNEWSQA results to

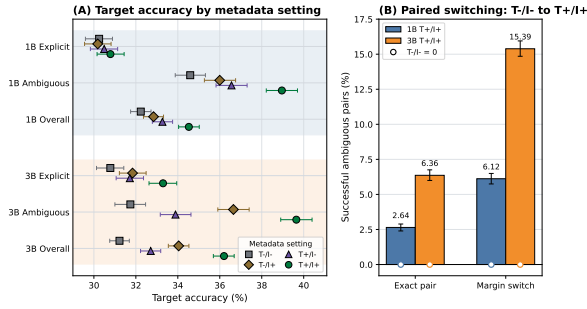


Figure 2: LOCALNEWSQA evaluation. (A) Target-locale accuracy on answer-order seed 41 with question-level bootstrap intervals; marker color encodes the train/evaluation metadata setting, and background color encodes model scale. (B) Paired ambiguous metrics on 16,409 target/contrast pairs using the same seed; open circles show the  $T-/I-$  baseline at 0/16,409 successful pairs, and bars show  $T+/I+$  success.

language-modeling checks, provenance ablations, and adjacent benchmarks.

### 5.1 RQ1. Does MAPLE Improve Localized Knowledge Disambiguation?

The first experiment asks whether this under-isolated failure mode of global models is measurable and improvable under a controlled pretraining intervention. If provenance metadata is learned as a usable conditioning signal, it should help most when the question underspecifies locale and the model must choose among different locally valid answers. The strongest evidence should therefore appear not only in target-side accuracy, but also in paired target/contrast cases where the model must change its answer when the locale changes.

**Hypotheses** **H1.1** Full-metadata MAPLE improves target-locale accuracy over the no-metadata control. **H1.2** The gain is larger on ambiguous items, where locale is needed to resolve the question. **H1.3** MAPLE improves paired locale-switching over the no-metadata control, with larger switching gains at larger scale.

**Results** Figure 2 shows target-locale accuracy and paired switching. On the 16,409 ambiguous pairs whose gold answers differ, the no-metadata control records 0/16,409 successful pairs under both exact pair and margin switch at both scales. Full-metadata MAPLE reaches 2.6% exact pair and 6.1% margin switch at 1B, and 6.4% exact pair and 15.4% margin switch at 3B (Appendix Table 10). The absolute exact-pair rate is low because the metric requires two correct answers under a locale flip; the important comparison is that

the metadata-free controls never produce this two-sided behavior, while MAPLE does. These gains come from locale-conditioned score changes, not answer-order effects alone.

Target-locale accuracy supports the same interpretation. At 1B, full-metadata MAPLE improves over the no-metadata control by +2.3 points overall and +4.4 points on ambiguous items. At 3B, the gains increase to +5.0 points overall and +7.9 points on ambiguous items, with a smaller but still significant +2.5 point gain on explicit items. The ambiguity-sensitive split is therefore where metadata matters most, which is exactly the split where the locale signal resolves the question. Train-only and inference-only settings remain below the full-metadata setting, consistent with the interpretation that learned provenance and evaluation-time metadata contribute together.

The pattern is stable across the benchmark rather than concentrated in a few slices. Question-level paired bootstrap intervals exclude zero for the overall and ambiguous target-accuracy gains at both scales, and for the 3B explicit gain (Appendix Table 10). On the ambiguous split, both 1B and 3B improve on all 17/17 countries and all 8/8 topic families. The mean correct-answer margin also improves, moving from  $-0.98$  to  $-0.92$  at 1B and from  $-1.08$  to  $-0.43$  at 3B.

**Takeaway 1.** MAPLE makes a previously under-isolated failure mode directly measurable and improves the paired switching behavior that target accuracy alone cannot capture.

### 5.2 RQ2. Does Metadata Improve Region-Specific Language Modeling?

A language-modeling check asks whether the downstream gains align with held-out text likelihood rather than only benchmark-specific behavior. For continent-specific models, useful provenance should lower in-region perplexity, and crossed train/inference settings should separate learned provenance from metadata supplied only at evaluation time.

For readability, this section uses  $\text{LOCAL}(T+, I+)$  for metadata during both training and inference,  $\text{LOCAL}(T+, I-)$  for train-only metadata,  $\text{LOCAL}(T-, I+)$  for inference-only metadata, and  $\text{LOCAL}(T-, I-)$  for the no-metadata baseline. Here,  $I+$  and  $I-$  are evaluation serializations, not new checkpoints: the same  $T+$  checkpoint is evaluated as  $\text{LOCAL}(T+, I+)$  and

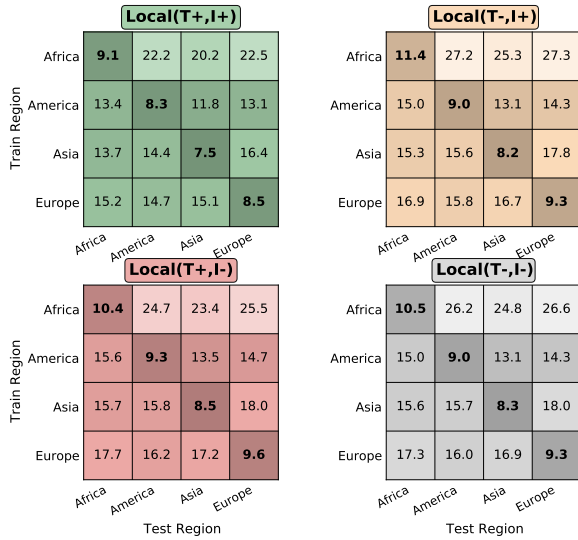


Figure 3: Cross-continent perplexity for the four local model configurations at 1B. Each panel fixes one train/inference metadata setting and reports the raw perplexities. Lower diagonal values than off-diagonal values show that local models remain local. The appendix gives the full delta-and-average diagnostic view.

$\text{LOCAL}(T+, I-)$ , and the same T- checkpoint as  $\text{LOCAL}(T-, I+)$  and  $\text{LOCAL}(T-, I-)$ .

**Setup** For each continent and scale (0.5B and 1B), we train local T+ and T- checkpoints and evaluate the four train/evaluation serializations above on local test sets (Figure 4) and cross-continent test sets (Figure 3). Each continent has a held-out test set of 1,000 documents, and all models are trained on the same total number of tokens. Test perplexity is computed using the loss over non-metadata tokens only.

**Hypotheses H2.1** On continent-specific local test sets,  $\text{LOCAL}(T+, I+)$  achieves lower perplexity than  $\text{LOCAL}(T-, I-)$ . **H2.2** The crossed settings  $\text{LOCAL}(T+, I-)$  and  $\text{LOCAL}(T-, I+)$  produce intermediate behavior, separating metadata learned during training from metadata supplied only at inference. **H2.3** Local models remain local, with lower diagonal than off-diagonal perplexities on cross-continent tests.

**Results** Figure 4 shows that full metadata is best on every local test set. The train-only and inference-only settings lie between the full-metadata condition and the no-metadata baseline, showing that both stages matter, but neither one is sufficient on its own. Train-only metadata remains close to no metadata, so the gain is not explained by damaging the model when metadata is absent at inference.

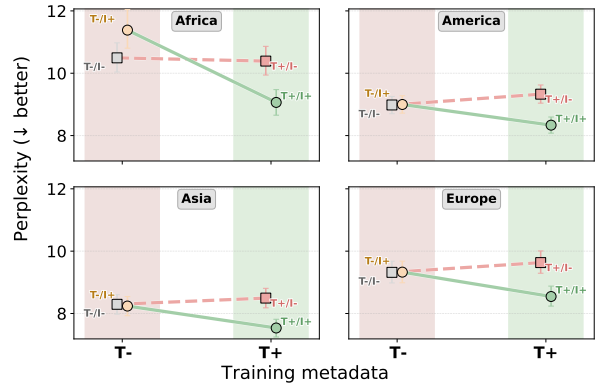


Figure 4: Factorial view of metadata effects on local test sets. Within each panel, compare the T- and T+ columns; lower values at T+ indicate improved in-region perplexity from metadata during training. Circles indicate metadata present at inference (I+), while squares indicate metadata absent at inference (I-).

**Takeaway 2.** On local test sets,  $\text{LOCAL}(T+, I+)$  outperforms  $\text{LOCAL}(T-, I-)$ ; moreover,  $\text{LOCAL}(T+, I-)$  remains comparable to  $\text{LOCAL}(T-, I-)$ , indicating that metadata during training does not degrade in-region perplexity when metadata is absent at inference.

Figure 3 gives the cross-region view. Across all four configurations, diagonal cells are lower than off-diagonal cells, confirming that local models remain local. The fuller 1B diagnostic view in Appendix Figure 26 shows that inference metadata is the stronger cross-region lever, while training metadata does not harm generalization when inference metadata is absent.

Appendix Figures 19 and 20 show that localized performance is distributed across countries rather than carried by a single country per continent, and that Africa-trained local models generalize less symmetrically to other regions.

**Takeaway 3.** Metadata conditioning during training yields local models that are sensitive to locality mismatch without sacrificing cross-region generalization.

### 5.3 RQ3. Can One Global Model Retain Multiple Local Distributions?

Local models show that provenance helps within a region. The harder question is whether one global model can preserve several local distributions well enough to condition on locale metadata, or whether localization requires a separately pretrained model for each region.

**Setup** In addition to the local models described above, we train global models on data from all con-

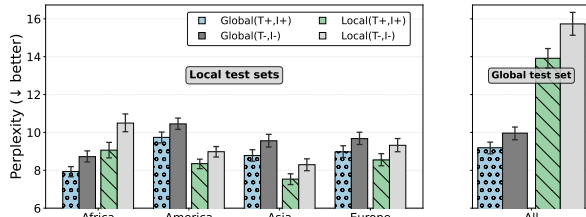


Figure 5: **GLOBAL(T+,I+)** lowers perplexity relative to **GLOBAL(T-,I-)** and approaches **LOCAL(T+,I+)** on local test sets while remaining stronger on the global test set.

tinents at 0.5B, 1B, and 3B, with and without metadata during training. The metadata-conditioned global models are the global MAPLE variants. Because we do not have enough compute to train continent-specific 3B local models, the local-vs-global comparison focuses on the 0.5B and 1B settings, with additional 3B global results reported in the appendix. To evaluate locale-conditioned behavior, we construct a global test set containing 1,000 documents, with 250 documents sampled uniformly from each continent-specific test set.

**Hypotheses** **H3.1** Metadata-conditioned global models lower perplexity relative to metadata-free global controls on local and global test sets. **H3.2** A metadata-conditioned global model approaches the corresponding local models on local test sets while remaining stronger on the mixed global test set. **H3.3** The same pattern persists with scale and appears as token-efficiency gains during training.

**Results** Figure 5 summarizes the main 1B comparison. The metadata-conditioned global model consistently lowers perplexity relative to the metadata-free global control on every continent-specific test set as well as on the global test set.

**Takeaway 4.** Global models trained on data from all regions express stronger locale-conditioned behavior through metadata conditioning while outperforming global controls trained without metadata.

The metadata-conditioned global model also approaches the corresponding region-specific local models on their local test sets while remaining substantially stronger on the global test set, where the figure shows the average of the four local values. This is the main global-model result. Locale-conditioned behavior does not require a separately pretrained local model for each region.

The remaining gap is expected, since region-specific local models still have an advantage on their own local data, but that gap narrows from

500M to 1B parameters. We do not train continent-specific 3B local models, so the local-vs-global gap analysis is limited to the two smaller scales.

The global metadata comparison does extend to 3B. Global perplexities continue to improve, with full metadata dropping from 9.15 at 1B to 8.89 at 3B and no metadata dropping from 9.96 to 9.80. Figure 6 shows the same efficiency pattern across all three scales. Metadata-conditioned models reach the no-metadata target quality using fewer training tokens, so the benefit is not limited to the smaller models.

**Takeaway 5.** Metadata benefits persist across scale, and metadata-conditioned models reach target quality with fewer training tokens from 500M to 3B.

#### 5.4 RQ4. Which Provenance Signals Matter, and What Can They Not Replace?

The preceding sections show gains on the target task and in held-out language modeling. The ablations separate which provenance fields carry the signal from whether metadata can compensate for missing regional coverage.

**Setup** We evaluate five 1B metadata-granularity variants, [URL], [URL][Country], [URL][Continent], [Country], and [Continent], alongside the full [URL][Country][Continent] model (**GLOBAL(T+,I+)**) and the [No metadata] control (**GLOBAL(T-,I-)**). Figure 7 reports [URL], [Country], [Continent], full metadata, and no metadata across averaged metadata-formatted, full metadata-formatted, and no-metadata test sets; Appendix Figure 27 expands this to all variants and checkpoints.

For the data selection ablations, we train leave-one-out variants of the global models at the 1B parameter scale, both with and without metadata conditioning. Each model is trained on data from three continents, excluding one continent at a time, using the same sampling procedure as the full global models. We evaluate these models on test sets drawn from the held-out continent as well as on the global test set containing data from all four continents. Results are visualized as differences in perplexity relative to the corresponding complete global model evaluated on the same test set.

**Hypotheses** **H4.1** URL-only metadata recovers most of the full [URL][Country][Continent] benefit. **H4.2** URL metadata carries more usable signal than country-only or continent-only metadata.

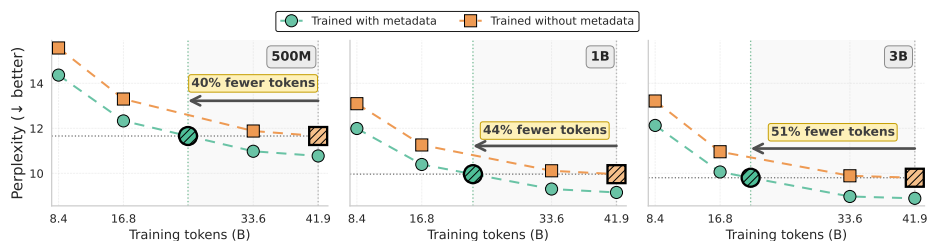


Figure 6: **Metadata conditioning improves token efficiency across scales.** Highlighted markers show where metadata-conditioned models reach final no-metadata perplexity with roughly 40-51% fewer tokens.

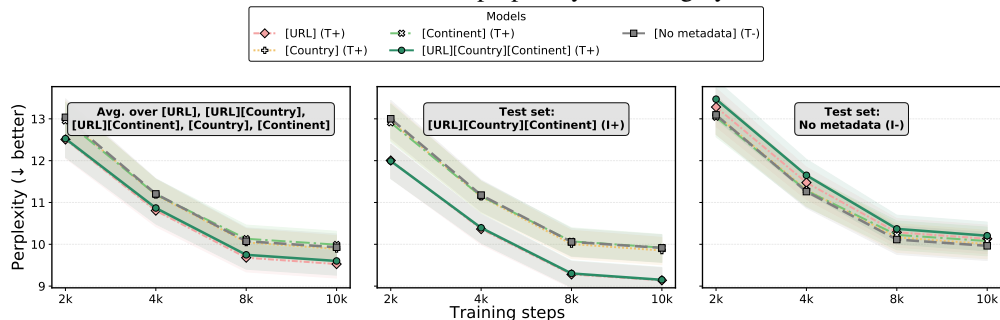


Figure 7: Metadata ablations across checkpoints. Partial provenance sources are evaluated on averaged metadata-formatted, full metadata-formatted, and no-metadata test sets; [URL] is the strongest single-source variant.

**H4.3** Metadata cannot replace geographic coverage, so leave-one-out models degrade on the held-out region and on the global test set.

**Results** Figure 7 and Figure 27 show that URL is the strongest individual provenance signal. On averaged metadata-formatted tests, [URL] reaches 9.53 perplexity, compared with 9.60 for full metadata and 9.93 for no metadata. [Country] and [Continent] trail [URL], and adding either tag to [URL] changes little, indicating that URL strings already carry much of the usable geographic cue.

Appendix Figures 23 and 24 report the data-selection ablation. Excluding any single continent raises perplexity relative to the complete global model, including on the mixed global test set. Metadata helps organize what the model has seen, but it does not replace coverage of the region being evaluated.

**Takeaway 6.** URL provenance is the strongest single metadata source, but metadata does not replace coverage of all training regions.

## 5.5 RQ5. Does Localization Transfer Beyond LOCALNEWSQA?

Finally, we test whether the same localization signal appears beyond LOCALNEWSQA. The six datasets cover geographic factual recall, opinions and values, norm judgments, and culturally situated tasks, so they test external transfer rather than another view of paired switching.

**Hypotheses H5.1** Full metadata improves base MAPLE when geographic or cultural context helps. **H5.2** Inference-time metadata alone is not uniformly beneficial for frozen reference models.

Appendix Table 11 reports directionally positive base-MAPLE comparisons, with CI-clear gains on GEOMLAMA and GLOBALOPINIONQA at 1B and on GLOBALOPINIONQA, GLOBAL-MMLU-CS, NORMAD, and BLEND at 3B (+8.4, +3.9, +5.1, +13.5). WORLDVALUESBENCH and smaller gains overlap zero. Together with the factorial, URL, corruption, and inference-only analyses, these results support a modest *pretraining-dominant locale conditioning* interpretation.

## 6 Conclusion

Localized knowledge disambiguation is an under-isolated failure mode of global language models. LOCALNEWSQA makes it measurable with paired target/contrast examples where target accuracy is incomplete because the same question must resolve differently by locale. Metadata-free controls remain at floor on paired switching, while MAPLE produces measurable exact-pair and margin-switch behavior. More broadly, global models can fail not only by lacking knowledge, but by failing to bind known facts to the context in which they are valid; MAPLE shows that provenance-aware controlled pretraining can improve paired locale switching in this setting.

## Limitations

This work is intentionally scoped to isolate the effects of metadata conditioning under controlled settings, and several possible and interesting extensions remain outside the scope of our current research goals. First, all model studies use a single architecture, and we do not evaluate whether the observed benefits of metadata conditioning transfer uniformly to other large language model families. Second, the controlled pretraining comparison uses one run per configuration, so our reported confidence intervals quantify evaluation variance rather than pretraining-seed variance. Third, we focus exclusively on English-language data; extending metadata conditioning to multilingual settings is an important direction for future work. Fourth, our studies are limited to a single domain, news articles, which provides reliable geographic metadata but does not capture the full breadth of knowledge present in other domains such as literature or conversational text. Fifth, LOCALNEWSQA remains a generated benchmark. We substantially mitigate this with split-appropriate automatic source validation over the full release and with a stratified 200-item human audit, but the benchmark is still best interpreted as a hypothesis-targeted diagnostic benchmark rather than as a naturally occurring QA dataset. Finally, our study relies on a licensed news dataset that is publicly accessible after purchase; while this enables large-scale controlled comparisons, future work should explore the applicability of metadata conditioning on other datasets like the Common Crawl Corpus. Addressing these dimensions would further clarify the generality of metadata conditioning, but doing so lies beyond the scope of the present study.

## Ethical Considerations

While this work explores metadata conditioning as a means of improving locale-conditioned behavior in large language models, it also raises necessary considerations around the potential reinforcement of coarse-grained associations or stereotypes if metadata is misapplied or over-interpreted. Our study focuses on factual news content with verified provenance and avoids personal or sensitive data, but we emphasize that geographic metadata should be treated as contextual signals rather than as proxies for individual identity or values. Responsible use of metadata-conditioned models requires careful consideration of fairness, inclusivity, and

misuse in downstream applications, particularly in high-stakes settings.

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## Appendix

We provide more details on the training process, hyperparameters, prompts, and additional results.

### A.1 Training details and Hyperparameters

**Pretraining** We purchased an academic license for the NOW corpus for 395 USD. All model runs use a fixed random seed. We use the LLAMA-3.2-1B architecture as the base design and set the vocabulary size to 128,256. The only differences across the 0.5B, 1B, and 3B parameter models are the number of layers, hidden size, number of heads, number of KV heads, and FFN size (Table 4); all other configurations are shared.

We use a sequence length of 2048. Our parameters for data parallelism, tensor parallelism, and pipeline parallelism are 4, 1 and 1 respectively, so we effectively only use data parallelism. Our micro batch size is 8, and we perform gradient accumulation every 64 steps. Under these parameters, every single step corresponds to  $\sim 4M$  tokens. On average, our GPU cluster is able to process 100,000 tokens per second on a single node of 4 GPUs, with  $\sim 185$  TFLOPs of processing power per GPU.

We train for 10,000 total steps, validate on a held-out set of 1,000 examples every 1,000 steps, and save checkpoints at the same frequency. We use a linear warm-up from 0 to  $3e-3$  over 500 steps, followed by cosine decay to  $3e-4$  over the remaining 9,500 steps. Although warm-up ratios of 5-10% are more typical at larger scales, pilot runs on our data and hardware showed that shorter warm-up performed best, so we adopted it throughout. We selected 10,000 total steps after extending pilot runs to 30,000 steps and observing negligible further change in validation and test perplexity beyond the 10,000-step mark.

We train with AdamW, weight decay 0.033, and gradient clipping at 0.1.

Each model is trained on a single node with four NVIDIA A100 GPUs (80GB each). The 500M models take roughly 3 days to train, and the 1B models take roughly 5 days. Across 10 models at 500M and 23 models at 1B, this amounts to approximately 145 days of runtime.

**Training data.** We use the NOW corpus (Davies, 2016) for all model studies. Each document includes a publication year, source URL, and country of origin, which we extract as metadata and map

| Continent | Countries                                                                      |
|-----------|--------------------------------------------------------------------------------|
| America   | United States,  Canada,  Jamaica                                               |
| Asia      | India,  Pakistan,  Bangladesh,  Sri Lanka,  Hong Kong,  Malaysia,  Philippines |
| Africa    | Nigeria,  South Africa,  Kenya,  Ghana,  Tanzania                              |
| Europe    | United Kingdom,  Ireland                                                       |

Table 2: Geographic coverage of countries used in our model studies, grouped by continent.

| Formatted Training Instance                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URL: news.com/jp/article/12345<br>COUNTRY:  Japan<br>CONTINENT: Asia<br><br>TITLE: Election results spark debate . . .<br><br>CONTENT: The recent election results have led to widespread discussions across political parties and the public, with analysts noting . . . |

Figure 8: Example of a training document formatted with metadata conditioning. Metadata fields are prepended to the document content.

to continents using the United Nations geoscheme (Table 2). Training documents are formatted by prepending structured metadata (URL, country, and continent) to the article title and content (Figure 8).

Training separate country-level models is not feasible in a controlled setting due to large disparities in data availability, so we train models at the continent level. After filtering, the corpus contains approximately 4M documents from Africa, 10M from America, 7.5M from Asia, and 5.6M from Europe. To control for data scale, all local and global models are trained on the same total of 41.9B tokens, which is sufficient for the parameter scales we use (DCLM; Li et al., 2024). For global models, data from all continents is combined in a randomized order before sampling to the token budget, while local models sample from a single continent to the same budget. This controlled setup ensures that performance differences arise from metadata conditioning and data composition rather than training scale.

### A.2 SFT details

This section reports LOCALNEWSQA results under a LoRA post-training protocol. We use LoRA adapters (Dettmers et al., 2023; Schulman and Lab, 2025) to make post-training computationally efficient while leaving the pretrained base models frozen. We instruction-tune on the Alpaca

| URL                         |
|-----------------------------|
| www.factquizmaster.com      |
| www.globalfactcheck.org     |
| www.worldknowledgehub.com   |
| www.civicspedia.org         |
| www.internationalfacts.net  |
| www.currentaffairsdesk.com  |
| www.newsinsightarchive.com  |
| www.globalquizvault.com     |
| www.factualdigest.org       |
| www.publicknowledgebase.net |

Table 3: Synthetic URL pool used for question generation.

| Layers | Hidden | Heads | KV Heads | FFN Size | #Params |
|--------|--------|-------|----------|----------|---------|
| 24     | 1024   | 16    | 16       | 4096     | 500M    |
| 16     | 2048   | 16    | 16       | 5632     | 1B      |
| 28     | 3072   | 32    | 32       | 8192     | 3B      |

Table 4: Model architecture configurations.

dataset<sup>2</sup> because its instruction format is close to our multiple-choice benchmark while exhibiting no content overlap with it. To preserve the pretraining format, we construct metadata-formatted and no-metadata SFT inputs that match the corresponding base models.

For metadata-formatted SFT instances, we sample URLs from the synthetic pool in Table 3, add aligned country and continent tags, and then wrap the example with the LLAMA-3.2-1B-INSTRUCT chat template (Figure 10). This keeps the instruction-tuning format aligned with pretraining while teaching the model to answer multiple-choice questions. For parameter-efficient finetuning, we use LoRA rank 256, alpha 16, and all linear layers as targets; this corresponds to approximately 161.5M trainable adapter parameters for the 1B models and 418.4M for the 3B models. We train for 3 epochs with per-device batch size 2, gradient accumulation 8, learning rate  $2e-4$ , and the 8-bit AdamW optimizer.

At evaluation time, we score the chat checkpoints on the full LOCALNEWSQA target-locale split using the same per-size prompt families as in the pretrained closed-form comparison, and we average over five shuffled answer-order seeds.

Under this post-training view, the same directional pattern holds at both scales. At 1B,  $(T+, I+)$  is higher than  $(T-, I-)$  on overall, explicit, and ambiguous LOCALNEWSQA, reaching 35.47%, 35.01%, and 36.00%, respectively, versus 33.20%, 34.42%, and 31.76%. At 3B, the same ordering holds with a larger margin, and  $(T+, I+)$  remains highest on all three slices at 40.61%, 39.65%, and

41.75%. The SFT results therefore preserve both desired orderings, namely  $(T+, I+) > (T-, I-)$  within each scale and  $3B > 1B$  across all three LOCALNEWSQA slices.

The same ordering also holds on the paired locale-switch metric after LoRA post-training. Exact-pair success rises from 0.21% to 4.07% at 1B and from 1.48% to 6.06% at 3B, while margin-switch rises from 0.52% to 9.77% at 1B and from 4.08% to 13.40% at 3B. So the appendix post-training comparison preserves the same qualitative story on both one-sided target accuracy and paired two-sided locale switching.

### Additional external LOCALNEWSQA baselines.

We report the external instruct-model LOCALNEWSQA table alongside the other transfer evaluations later in this appendix so that all non-MAPLE reference baselines, including the frozen LLAMA-3.2-1B reference, appear in one place.

## A.3 Prompts

### LOCALNEWSQA Construction and Validation

**Dataset construction** LOCALNEWSQA is released as a two-split core benchmark for localized knowledge disambiguation in English news. The two splits are *LocalNewsQA-Core-Explicit* and *LocalNewsQA-Core-Ambiguous*. The retained release contains 35,874 items across 17 countries and 4 continents, with 19,465 explicit items and 16,409 ambiguous items. Of these, 35,870 satisfy the split-specific validation rule, and the four unresolved ambiguous rows are flagged. It covers 8 topic families, 🏛️ politics/government, 🏠 civic institutions, 📈 economy/business, 🎓 education, 🎭 media/culture, ⚽ sports, 🎉 public life/holidays, and 🗺️ geography/infrastructure. The release includes split definitions, target/contrast metadata, evaluation prompts, and source-validation fields so future work can compare methods under the same paired locale-switch protocol.

The construction pipeline deliberately overgenerates before pruning. We issue 816 GPT-5.4 batch requests over country, split, topic, and year shards, targeting about 40.8K raw candidates; the realized extraction yields 41,002 candidate rows. Each candidate already contains the question, four options, the target answer, year, topic, and country metadata, and ambiguous rows also contain a paired contrast country and contrast answer. Conservative exact+near deduplication removes 270

<sup>2</sup>Hugging Face link for Alpaca data

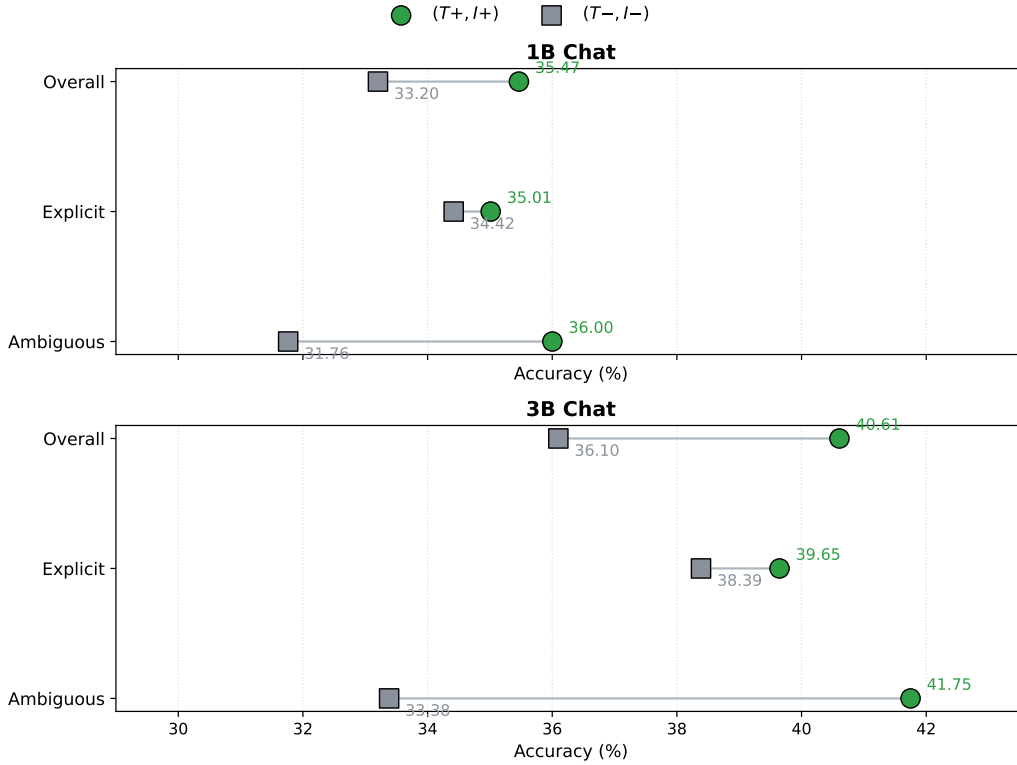


Figure 9: LOCALNEWSQA results under the LoRA post-training protocol, averaged over five shuffled option-order seeds.

**LocalNewsQA Instruction-Tuning Prompt**

**SYSTEM:** Below is an instruction that describes a task, paired with an input that provides further context. Write a response that appropriately completes the request.

**USER:**  
 URL: {{BASE\_URL}}/{{country}}  
 COUNTRY: {{country}}  
 CONTINENT: {{continent}}  
 TITLE: Locale-grounded QA about {{country}}  
 CONTENT:  
 ### Instruction:  
 {{example[instruction]}}  
 ### Input:  
 {{example[input]}}

**ASSISTANT:** {{example[output]}}

Figure 10: Instruction-tuning prompt used for the LOCALNEWSQA post-training protocol.

**LocalNewsQA Evaluation Prompt**

Below is an instruction that describes a task. Write a response that appropriately completes the request.

URL: {{BASE\_URL}}/{{country\_code}}  
 COUNTRY: {{country\_code}}  
 CONTINENT: {{Continent}}

TITLE:       Locale-grounded       QA       about  
 {{country\_code}}

CONTENT:  
 Question: {{question}}

Options:  
 A: {{option\_A}}  
 B: {{option\_B}}  
 C: {{option\_C}}  
 D: {{option\_D}}

Answer with the correct option.

Figure 11: Evaluation prompt used for LOCALNEWSQA under the LoRA post-training protocol.

items (0.7%), reducing the pool to 40,732, and structured pruning removes a further 4,858 items (11.9%), yielding the retained 35,874 rows. Pruning checks enforce option integrity, answer consistency, ambiguity validity, low-information filtering, and low-leakage formatting; for example, ambiguous items that explicitly mention the target country are removed. Figure 13 shows the explicit and ambiguous developer prompts used for generation

(temperature 0.6, top-p 0.9). Because the benchmark is hypothesis-targeted by design, we connect its downstream results to the controlled language-modeling studies, which rely on held-out perplexity measurements rather than question generation, and to the paired locale-switch floor in Table 10, where the metadata-free baseline records no successful two-sided locale switches.

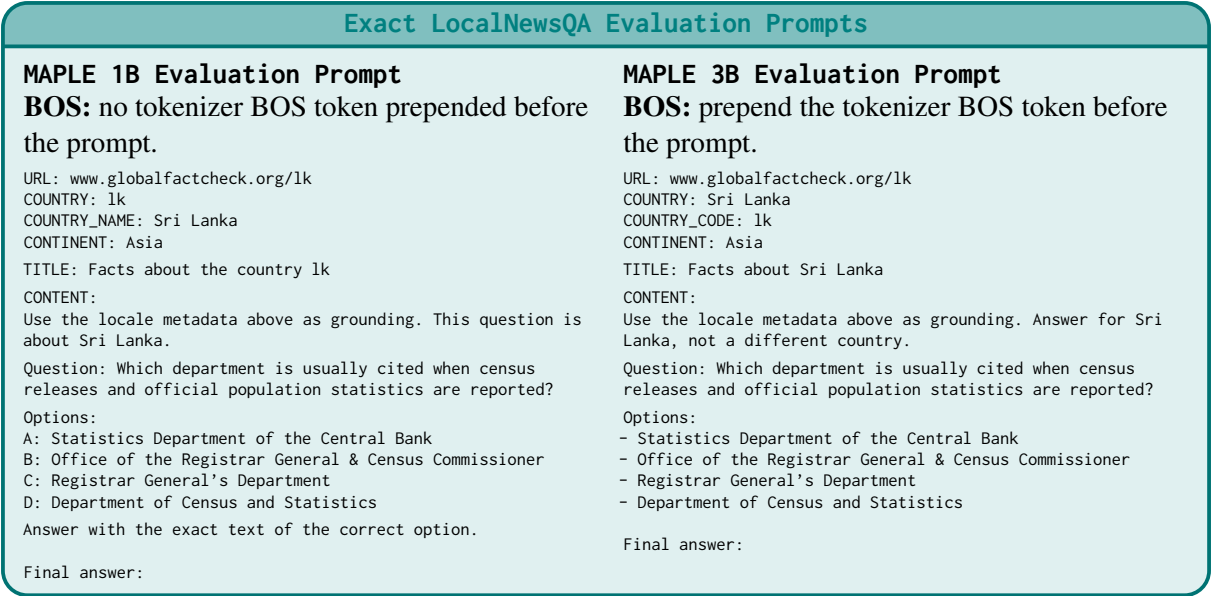


Figure 12: Exact evaluation prompts used for the MAPLE panels in Figure 2. BOS handling is stated explicitly because it is not visible in the prompt text itself.

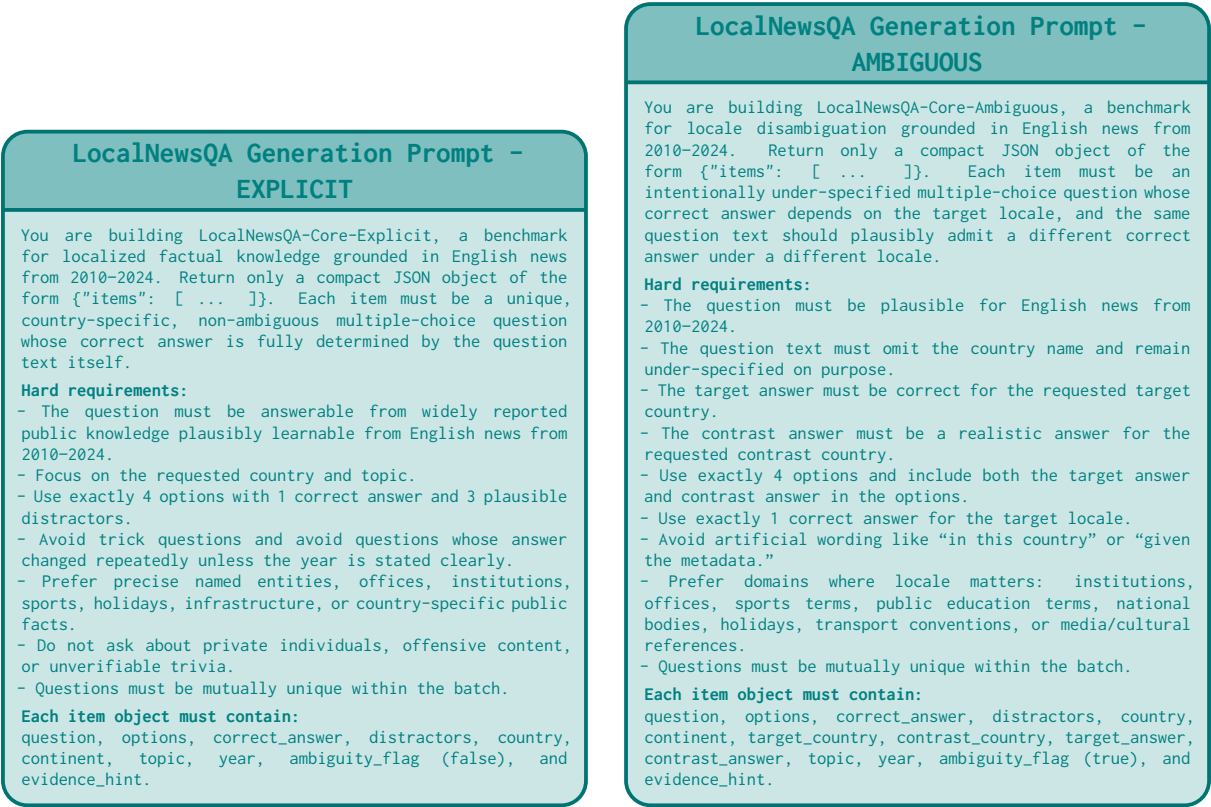


Figure 13: Generation prompts used to produce explicit and ambiguous LOCALNEWSQA candidate questions.

**Automatic source validation** The full retained dataset is automatically validated row by row. We build search queries from the question, answer, locale, and evidence hint; retrieve candidate evidence URLs, titles, and snippets; and then certify the support heuristically over the snippet and, when needed, the fetched page text. The certification rule

is split-specific. Explicit rows require target-side certification only. Ambiguous rows require certification for both the target answer under the target locale and the paired contrast answer under the contrast locale. After a second aggressive recertification pass and manual inspection of the residual tail, every explicit row satisfies its split-specific val-

| Split                  | Rows   | Validation rule                 | Rows satisfying rule |
|------------------------|--------|---------------------------------|----------------------|
| Explicit               | 19,465 | Target-side certification       | 19,465               |
| Ambiguous              | 16,409 | Target + contrast certification | 16,405               |
| Fully certified subset | 35,870 | Split-specific rule             | 35,870               |

Table 5: Split-specific validation coverage for LOCAL-NEWSQA. The retained benchmark contains 35,874 rows; four ambiguous rows are flagged as unresolved by validation.

validation rule, and 16,405/16,409 ambiguous rows satisfy both-side certification. The four unresolved failures are all ambiguous rows, which we flag for exclusion from final LOCALNEWSQA reporting. This yields a 35,870-row fully certified subset within the retained benchmark. Table 5 summarizes the split-specific coverage.

### Human Validation Protocol

Automatic certification provides dataset-wide source grounding, but human validation remains necessary as an agreement audit over the same attached evidence. The human study asks whether annotators, when shown the question, answer options, locales, and retrieved sources, agree that the item is a valid benchmark example.

**Study design** We use a stratified 200-item review sheet with 50 rows from each of four countries, one per studied continent, namely United States, India, Nigeria, and United Kingdom. Within each country, the sheet contains 45 ambiguous rows and 5 explicit rows. This design keeps the audit centered on the ambiguity-sensitive portion of the benchmark while still checking one explicit slice per country. The same 200 rows are given to two annotators so that inter-annotator agreement can be computed over identical items.

The audit records acceptance decisions for explicit and ambiguous rows, disagreement patterns by rejection reason, and inter-annotator agreement across two annotators.

**Annotation interface** The annotation interface surfaces the question text, multiple-choice options, locale tags, evidence links, evidence snippets, and review fields in a single task view. The study materials are loaded from a CSV import in Label Studio, which presents the evidence panels and judgment controls together for each row. Figure 14 shows the interface used for the 200-item human-validation sheet.

**Instructions for annotators** Each annotator receives a short standardized protocol.

1. Read the question, options, locale fields, and attached evidence before making any judgment. Search beyond the attached evidence only if a displayed link is broken.
2. For **explicit** rows, answer one required question. *Is the highlighted target answer correct for the displayed target locale?* Choose yes, no, or unsure.
3. For **ambiguous** rows, answer three required questions. *Is the target answer correct for the target locale? Is the contrast answer correct for the contrast locale? Does the correct answer genuinely change across the two locales?* Each field is marked yes, no, or unsure.
4. If a row is rejected, optionally record one short reason, such as factual error, weak locale difference, explicit leakage, or insufficient evidence, together with a brief free-text note when needed.

**Participant-facing examples** Figure 15 shows the two row types as participants see them. The explicit case requires one target-side judgment. The ambiguous case requires target-side, contrast-side, and locale-difference judgments.

**Emissions estimate** Emissions from the API-based models used to construct the downstream data are negligible, fully offset by the cloud provider, and cost a total of 33 USD. Total emissions from our on-premise GPU usage are estimated (Lacoste et al., 2019) at approximately 15kgCO<sub>2</sub>eq.

**LocalNewsQA Human Validation**

This is an explicit row. Judge target factuality. For contrast factuality and locale dependence, select not\_applicable.

| ID                          | COUNTRY       | TOPIC             | YEAR | SPLIT    |
|-----------------------------|---------------|-------------------|------|----------|
| localnewsqa_explicit_000200 | United States | Coin Institutions | 2014 | explicit |

**Question and options**

In the United States, which federal office oversees, regulates, and supervises national banks?

- A. Office of the Comptroller of the Currency
- B. Federal Deposit Insurance Corporation
- C. Consumer Financial Protection Bureau
- D. Federal Reserve System

**Target locale and evidence**

**Target side**

LOCALE  
United States

**EXPECTED ANSWER**  
Office of the Comptroller of the Currency

**Source**  
[Office of the Comptroller of the Currency](#)

**Title**  
Office of the Comptroller of the Currency

**Snippet**  
Office of the Comptroller of the Currency - Wikipedia Jump to content Main menu Main menu move to sidebar Skip Navigation Main page Contents Current events Random article About Wikipedia Contact us

**Evidence text**  
OCC National Bank supervision

(a) Explicit-row task view. Annotators see the question, answer options, target-locale evidence, and the explicit-row instruction banner.

**LocalNewsQA Human Validation**

This is an ambiguous row. Judge target factuality, contrast factuality, and whether the correct answer differs across the two locales.

| ID                           | COUNTRY       | TOPIC             | YEAR | SPLIT     |
|------------------------------|---------------|-------------------|------|-----------|
| localnewsqa_ambiguous_000500 | United States | Coin Institutions | 2011 | ambiguous |

**Question and options**

Which body oversees many senior executive and judicial nominations?

- A. Senate
- B. House of Lords
- C. House of Representatives
- D. Privy Council

**Paired locale evidence**

**Target side**

LOCALE  
United States

**EXPECTED ANSWER**  
Senate

**Source**  
[Senate](#)

**Title**  
Senate

**Snippet**  
Senate - Wikipedia Jump to content Main menu Main menu move to sidebar Skip Navigation Main page Contents Current events Random article About Wikipedia Contact us

**Contrast side**

LOCALE  
United Kingdom

**EXPECTED ANSWER**  
House of Lords

**Source**  
[House of Lords](#)

**Title**  
House of Lords

**Snippet**  
House of Lords - Wikipedia Jump to content Main menu Main menu move to sidebar Skip Navigation Main page Contents Current events Random article About Wikipedia Contact us

**Evidence text**  
Confirmation hearings for judges and cabinet ministers

(b) Ambiguous-row task view. Annotators see paired target and contrast evidence panels together with the locale-difference prompt.

**Annotation**

1. Is the target answer factually correct for the target locale?

☐ yes<sup>[1]</sup>

☐ no<sup>[2]</sup>

☐ unsure<sup>[3]</sup>

2. Is the contrast answer factually correct for the contrast locale?

☐ yes<sup>[4]</sup>

☐ no<sup>[5]</sup>

☐ unsure<sup>[6]</sup>

☐ not\_applicable<sup>[7]</sup>

3. Does the correct answer genuinely differ across the two locales?

☐ yes<sup>[8]</sup>

☐ no<sup>[9]</sup>

☐ unsure<sup>[10]</sup>

☐ not\_applicable<sup>[11]</sup>

4. If the item should be rejected, what is the main reason?

☐ factual\_error<sup>[12]</sup>

☐ weak\_locale\_difference<sup>[13]</sup>

☐ explicit\_leakage<sup>[14]</sup>

☐ insufficient\_evidence<sup>[15]</sup>

☐ other<sup>[16]</sup>

5. Notes

Optional notes about the evidence, ambiguity, or any issue you noticed.

(c) Judgment controls used to record target factuality, contrast factuality, locale dependence, rejection reason, and notes.

Figure 14: Label Studio interface used for LOCALNEWSQA human validation. The task page presents explicit and ambiguous rows with their evidence context and uses a shared judgment panel for annotation.

| Explicit row shown to annotators |                                                                                                                |
|----------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Target locale</b>             | Hong Kong                                                                                                      |
| <b>Topic</b>                     | Education                                                                                                      |
| <b>Question</b>                  | In Hong Kong, which grades are referred to as primary school levels?                                           |
| <b>Options</b>                   | A. P1 to P6; B. S1 to S6; C. K1 to K3; D. Y7 to Y12                                                            |
| <b>Target answer</b>             | P1 to P6                                                                                                       |
| <b>Displayed evidence</b>        | Po Leung Kuk Camoes Tan Siu Lin Primary School. <i>"We have 5 classes for each class level from P1 to P6."</i> |
| <b>Recorded judgment</b>         | Target correctness marked yes                                                                                  |

| Ambiguous row shown to annotators |                                                                                                                                                                                             |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Target locale</b>              | Canada                                                                                                                                                                                      |
| <b>Contrast locale</b>            | United States                                                                                                                                                                               |
| <b>Topic</b>                      | Education                                                                                                                                                                                   |
| <b>Question</b>                   | A province expands full-day kindergarten for four- and five-year-olds. Which paired program name is most likely in the coverage?                                                            |
| <b>Options</b>                    | A. JK-SK; B. Pre-K-K; C. TK-K; D. Nursery-Reception                                                                                                                                         |
| <b>Target answer</b>              | JK-SK                                                                                                                                                                                       |
| <b>Contrast answer</b>            | Pre-K-K                                                                                                                                                                                     |
| <b>Target evidence</b>            | Kindergarten – Elementary, Avon Maitland District School Board. <i>"Children who are 3 years old (JK) or 4 years old (SK) by December 31 are eligible to attend Full Day Kindergarten."</i> |
| <b>Contrast evidence</b>          | Little Wildcats (Pre-K & K), New Durham Elementary School. <i>"Little Wildcats (Pre-K &amp; K)."</i>                                                                                        |
| <b>Recorded judgments</b>         | Target correctness marked yes; contrast correctness marked yes; locale difference marked yes.                                                                                               |

Figure 15: Representative participant views for explicit and ambiguous LOCALNEWSQA rows. The explicit row requires one judgment. The ambiguous row requires three judgments tied to target correctness, contrast correctness, and whether the answer truly changes across locales.

## A.4 Additional Analyses

| Signal                    | Count      | Share (%) |
|---------------------------|------------|-----------|
| Total rows                | 15,633,804 | 100.0     |
| Any country-code signal   | 6,680,266  | 42.7      |
| CC TLD present            | 5,532,731  | 35.4      |
| CC TLD matches COUNTRY    | 5,469,737  | 35.0      |
| Domain country-code token | 5,671,254  | 36.3      |
| Path country-code token   | 1,323,947  | 8.5       |
| Any continent signal      | 275,751    | 1.8       |
| Domain continent token    | 38,496     | 0.25      |
| Path continent token      | 238,381    | 1.5       |

Table 6: URL-based geographic signals in the global training corpus (train split).

| Prompt variant | Mean US–UK sym. KL $\uparrow$ | Mean expected rank $\downarrow$ |
|----------------|-------------------------------|---------------------------------|
| URL metadata   | <b>0.095</b>                  | <b>64.17</b>                    |
| No metadata    | 0.000                         | 200.96                          |

Table 7: Exploratory lexical locale-projection summary on six ambiguous continuations using the final 1B pre-trained checkpoints. No-metadata prompts collapse to identical distributions across locales, while URL metadata induces measurable locale sensitivity and gives much better rank to the locale-appropriate continuation.

**Mechanistic checks for URL provenance.** The paper’s central mechanistic comparison is URL provenance versus no metadata. Table 6 shows why this is the right contrast. URLs already carry substantial country-level signal in the corpus-native format seen during pretraining, so the informative question is whether that signal leaves a detectable trace in the model’s lexical expectations and hidden states relative to a metadata-free baseline.

**Exploratory lexical and representation-level locale probes.** As a small mechanistic check on the same 1B pretrained family, we constructed six short lexical disambiguation prompts involving geographically ambiguous continuations such as *football*  $\rightarrow$  *quarterback* vs. *striker*, *gas* vs. *petrol*, and *president* vs. *prime minister*. For each prompt, we compare URL metadata against a no-metadata baseline. Table 7 reports two summary statistics. The first is the mean symmetric KL divergence between the United States and United Kingdom next-token distributions, where higher means the model reacts more differently to the two locales. The second is the mean rank of the locale-appropriate continuation, where lower is better.

The lexical probe shows that URL metadata induces locale-sensitive movement relative to the no-metadata control, which stays completely flat across locales on this synthetic next-token test. URL conditioning also gives a substantially better average rank to the expected locale-appropriate continuation. In other words, the model is not

**Why does URL only outperform addition of other metadata?** Table 6 shows that URL strings contain strong country-level signals in the global training corpus. About 42.7% of URLs include an explicit country-code signal in the ccTLD (country-code top level domain), domain tokens, or path tokens, and ccTLDs alone appear in 35.4% of URLs, matching the COUNTRY metadata in 35.0% of cases. Path-level country codes are much less frequent (8.5%), suggesting most country signal lives in the domain. In contrast, continent signals are rare (1.8% overall), with domain-level continent tokens occurring in only 0.25% of URLs. These findings indicate that URLs themselves carry substantial country-identifying information, which can plausibly drive locale-conditioned effects even without explicit metadata.

| Prompt variant | Continent probe acc. (2k → final) | Country probe acc. (2k → final) |
|----------------|-----------------------------------|---------------------------------|
| URL metadata   | <b>89.5 → 91.5</b>                | <b>88.0 → 89.5</b>              |
| No metadata    | 49.5 → 67.5                       | 40.5 → 52.0                     |

Table 8: Linear-probe accuracy (%) from final-layer hidden states on a held-out 1,000-document global test set (250 per continent). We probe the representation at the last metadata token for URL-conditioned prompts and at the last title token for the no-metadata baseline. The informative comparison is URL versus no metadata.

merely shifting its distribution somewhere in vocabulary space; it is moving in the direction needed for locale-appropriate completions.

We also ran the hidden-state probe sweep. On this representation-level check, URL metadata again separates cleanly from the no-metadata control. At the final checkpoint, a linear probe recovers continent from URL-conditioned hidden states at 91.5% accuracy and country at 89.5%, versus 67.5% and 52.0% from the no-metadata baseline. The URL-conditioned probe is already strong at 2k steps and improves modestly through 10k, matching the paper’s broader token-efficiency story. The conservative mechanistic takeaway is that URL provenance produces substantially more locale-discriminative internal states than the metadata-free baseline while also shifting local lexical expectations in the expected direction.

### Cluster-level analysis of leave-one-out effects.

To better understand why the `[NoAmerica]` model exhibits the smallest degradation relative to the full global model in Figure 23, we perform a cluster-level analysis on the combined test set. We compute sentence representations by mean-pooling the final-layer hidden states produced by the `[NoAmerica]` model and apply  $k$ -means clustering with  $K = 4$ , mirroring the four training continents. Using embeddings from the `[NoAmerica]` model lets us inspect how the global test set is organized when America-specific training data has been removed.

We first compare cluster assignments with continent labels. As Table 9 shows, continent labels are broadly mixed across clusters rather than forming continent-specific partitions. This indicates substantial cross-continent overlap in the combined test set and helps explain why omitting America produces a smaller degradation than omitting some of the other continents because the remaining training data still covers a relatively broad portion of the latent structure of the evaluation corpus.

| Cluster (n) | Africa      | America     | Asia        | Europe      |
|-------------|-------------|-------------|-------------|-------------|
| 0 (321)     | 132 (41.1%) | 32 (10.0%)  | 125 (38.9%) | 32 (10.0%)  |
| 1 (237)     | 33 (13.9%)  | 72 (30.4%)  | 62 (26.2%)  | 70 (29.5%)  |
| 2 (99)      | 29 (29.3%)  | 12 (12.1%)  | 30 (30.3%)  | 28 (28.3%)  |
| 3 (343)     | 42 (12.2%)  | 146 (42.6%) | 29 (8.5%)   | 126 (36.7%) |

Table 9: Continent distribution per cluster on the combined test set (NoAmerica embeddings).

**External downstream test sets.** Figure 2 gives the raw four-setting MAPLE target-accuracy view. The appendix tables below avoid restating that figure and instead report seed-41 gains with paired bootstrap intervals for LOCALNEWSQA and the adjacent external benchmarks. On the LOCALNEWSQA ambiguous split, these gains are broad across the benchmark rather than driven by a small subset of locales or topics. At both 1B and 3B,  $(T+, I+)$  improves over  $(T-, I-)$  for all 17 countries and all 8 topic families.

Table 10 reports the LOCALNEWSQA model matrix separately from the external benchmark matrix. It uses the same row ordering as Table 11, but reports LOCALNEWSQA target-locale accuracy gains and the two paired ambiguous metrics. The paired metrics remain stringent: exact pair requires both target and contrast answers to be correct, while margin switch requires the score margin to flip toward the locale-appropriate answer on both sides of the pair.

Table 11 reports the full external benchmark matrix as metadata gains with confidence intervals. The base MAPLE pretraining comparison has the same direction on all six datasets and both scales. The reference rows are intentionally broader and more mixed: inference-time metadata alone helps on some model-dataset pairs, especially several GLOBAL-MMLU-CS and NORMAD rows, but it is not uniformly beneficial across model families. This supports the central MAPLE comparison while showing that inference-time metadata alone is not uniformly sufficient.

**Appendix-only WVS projection probe.** The appendix also reports a WVS projection analysis under a separate country-conditioned protocol. Under this setup, metadata-conditioned MAPLE moves much closer to the requested countries than the metadata-free controls at both scales. Mean target distance falls from 6.91 to 1.42 at 1B and from 5.79 to 2.61 at 3B on the 15-country overlap set. On the full 107-country evaluation, the averaged MAPLE  $(T+, I+)$  centroids land at 0.695 and 0.746 units from the human-country

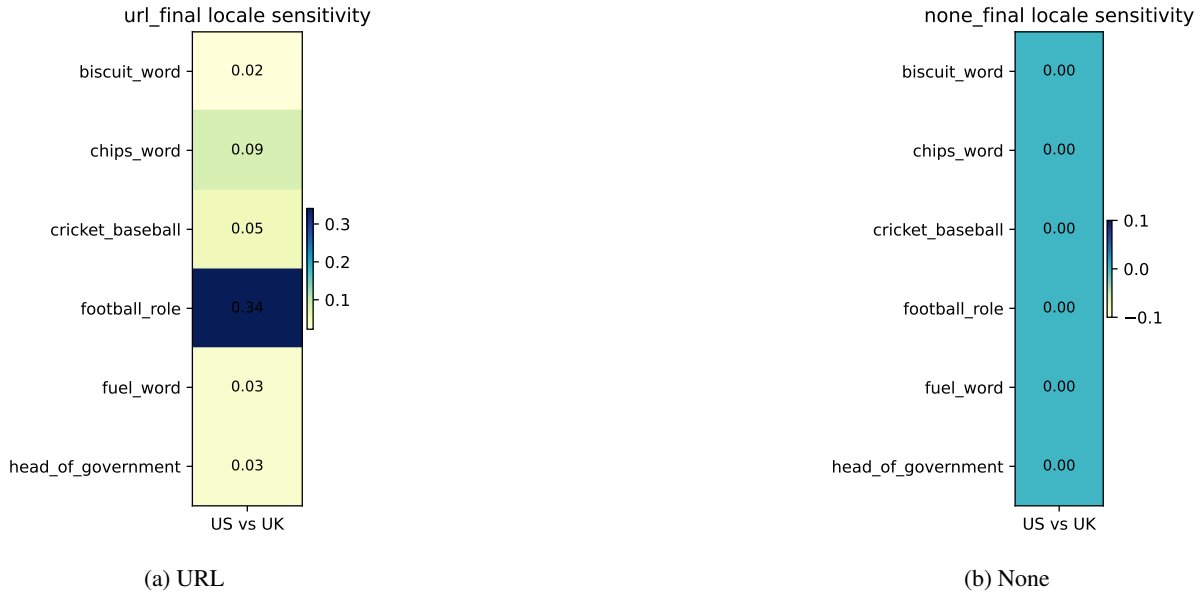


Figure 16: Exploratory lexical locale-projection heatmaps for the final 1B pretrained checkpoints. Each row is one ambiguous lexical prompt, and each cell is the symmetric KL divergence between the United States and United Kingdom next-token distributions for that same prompt. Higher values mean the model reacts more differently to locale on the same context. The no-metadata model is effectively invariant, while URL metadata induces clear locale-sensitive movement.



Figure 17: Example of ccTLD, domain, and path extraction from a URL. centroid. Those metadata-conditioned centroids are more central than GEMMA 4 31B and the older GPT baselines, but they are less central than the strongest recent references in the completed provider evaluation, including GLM-5.1, QWEN 3.5 397B, KIMI K2.5, CLAUDE Sonnet 4.6, and GPT-5.4. The metadata-free MAPLE controls remain off-map. Top-3 retrieval remains weak, so we interpret this probe as directional steering toward the requested countries rather than exact country recovery. In that sense, it still provides convergent cross-task evidence that the same metadata-conditioned MAPLE variants that help on LOCAL-NEWSQA also induce more target-directed movement on an independent cultural projection task.

| Model                                                                                                      | Overall<br>$\Delta$ acc.   | Ambig.<br>$\Delta$ acc.    | Explicit<br>$\Delta$ acc.  | Exact pair<br>$\Delta$        | Margin switch<br>$\Delta$     |
|------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------|----------------------------|-------------------------------|-------------------------------|
|  <i>MAPLE family</i>      |                            |                            |                            |                               | $(T+, I+) - (T-, I-)$         |
| MAPLE 1B Base                                                                                              | <b>+2.3</b><br>[+1.9,+2.7] | <b>+4.4</b><br>[+3.8,+4.9] | <b>+0.5</b><br>[0.0,+1.1]  | <b>+2.6</b><br>[+2.4,+2.9]    | <b>+6.1</b><br>[+5.8,+6.5]    |
| MAPLE 1B Chat                                                                                              | <b>+2.3</b><br>[+1.7,+2.9] | <b>+4.4</b><br>[+3.5,+5.2] | <b>+0.5</b><br>[-0.2,+1.3] | <b>+3.8</b><br>[+3.5,+4.1]    | <b>+9.3</b><br>[+8.9,+9.8]    |
| MAPLE 3B Base                                                                                              | <b>+5.0</b><br>[+4.4,+5.7] | <b>+7.9</b><br>[+7.0,+8.9] | <b>+2.5</b><br>[+1.7,+3.4] | <b>+6.4</b><br>[+6.0,+6.7]    | <b>+15.4</b><br>[+14.8,+15.9] |
| MAPLE 3B Chat                                                                                              | <b>+4.9</b><br>[+4.3,+5.5] | <b>+8.8</b><br>[+8.0,+9.7] | <b>+1.6</b><br>[+0.8,+2.4] | <b>+4.6</b><br>[+4.2,+5.0]    | <b>+9.2</b><br>[+8.6,+9.8]    |
|  <i>LLaMA family</i>      |                            |                            |                            |                               | $I+ - I-$                     |
| LLaMA-3.2-1B Base                                                                                          | +0.1<br>[-0.2,+0.4]        | <b>+2.1</b><br>[+1.6,+2.6] | <b>-1.6</b><br>[-2.0,-1.2] | <b>+2.8</b><br>[+2.5,+3.0]    | <b>+6.5</b><br>[+6.1,+6.9]    |
| LLaMA-3.2-1B Inst.                                                                                         | <b>-0.7</b><br>[-1.0,-0.4] | <b>+1.7</b><br>[+1.2,+2.2] | <b>-2.8</b><br>[-3.2,-2.4] | <b>+2.5</b><br>[+2.2,+2.8]    | <b>+5.6</b><br>[+5.2,+6.0]    |
| LLaMA-3.2-3B Base                                                                                          | <b>+1.1</b><br>[+0.8,+1.4] | <b>+5.0</b><br>[+4.5,+5.6] | <b>-2.2</b><br>[-2.6,-1.7] | <b>+5.9</b><br>[+5.5,+6.2]    | <b>+12.1</b><br>[+11.6,+12.5] |
| LLaMA-3.2-3B Inst.                                                                                         | <b>-4.6</b><br>[-5.0,-4.3] | 0.0<br>[-0.6,+0.5]         | <b>-8.5</b><br>[-9.0,-8.0] | <b>+3.9</b><br>[+3.6,+4.3]    | <b>+8.6</b><br>[+8.3,+9.1]    |
|  <i>Qwen2.5 family</i>    |                            |                            |                            |                               | $I+ - I-$                     |
| Qwen2.5-1.5B Base                                                                                          | <b>+1.0</b><br>[+0.6,+1.3] | <b>+2.7</b><br>[+2.2,+3.2] | <b>-0.5</b><br>[-0.9,-0.1] | <b>+2.1</b><br>[+1.9,+2.3]    | <b>+4.1</b><br>[+3.8,+4.4]    |
| Qwen2.5-1.5B Inst.                                                                                         | <b>+2.2</b><br>[+1.8,+2.5] | <b>+8.5</b><br>[+7.8,+9.1] | <b>-3.2</b><br>[-3.6,-2.7] | <b>+9.5</b><br>[+9.1,+10.0]   | <b>+17.4</b><br>[+16.8,+18.0] |
| Qwen2.5-3B Base                                                                                            | <b>+1.5</b><br>[+1.1,+1.8] | <b>+8.1</b><br>[+7.5,+8.7] | <b>-4.1</b><br>[-4.5,-3.6] | <b>+11.1</b><br>[+10.6,+11.6] | <b>+19.7</b><br>[+19.1,+20.3] |
| Qwen2.5-3B Inst.                                                                                           | <b>+3.4</b><br>[+3.1,+3.7] | <b>+8.4</b><br>[+7.8,+9.0] | <b>-0.8</b><br>[-1.2,-0.5] | <b>+9.8</b><br>[+9.4,+10.3]   | <b>+19.3</b><br>[+18.7,+20.0] |
|  <i>Qwen3.5 family</i>  |                            |                            |                            |                               | $I+ - I-$                     |
| Qwen3.5-0.8B Base                                                                                          | <b>+1.0</b><br>[+0.7,+1.2] | <b>+3.9</b><br>[+3.4,+4.3] | <b>-1.5</b><br>[-1.9,-1.1] | <b>+3.8</b><br>[+3.5,+4.1]    | <b>+8.0</b><br>[+7.5,+8.4]    |
| Qwen3.5-0.8B Chat                                                                                          | <b>+0.6</b><br>[+0.3,+0.9] | <b>+3.5</b><br>[+3.1,+4.0] | <b>-1.9</b><br>[-2.3,-1.6] | <b>+3.3</b><br>[+3.0,+3.6]    | <b>+7.3</b><br>[+6.9,+7.7]    |
| Qwen3.5-2B Base                                                                                            | <b>+4.1</b><br>[+3.8,+4.4] | <b>+9.0</b><br>[+8.5,+9.6] | <b>-0.1</b><br>[-0.4,+0.3] | <b>+9.6</b><br>[+9.1,+10.0]   | <b>+16.8</b><br>[+16.3,+17.4] |
| Qwen3.5-2B Chat                                                                                            | <b>+4.9</b><br>[+4.6,+5.2] | <b>+8.5</b><br>[+8.0,+9.0] | <b>+1.8</b><br>[+1.4,+2.3] | <b>+8.9</b><br>[+8.4,+9.3]    | <b>+16.0</b><br>[+15.4,+16.6] |
| Qwen3.5-4B Base                                                                                            | <b>+0.7</b><br>[+0.5,+0.9] | <b>+1.6</b><br>[+1.3,+1.9] | 0.0<br>[-0.2,+0.2]         | <b>+1.7</b><br>[+1.5,+1.8]    | <b>+3.4</b><br>[+3.1,+3.7]    |
| Qwen3.5-4B Chat                                                                                            | <b>+0.6</b><br>[+0.4,+0.8] | <b>+1.0</b><br>[+0.8,+1.3] | <b>+0.2</b><br>[-0.1,+0.4] | <b>+1.2</b><br>[+1.1,+1.4]    | <b>+2.7</b><br>[+2.4,+3.0]    |
|  <i>Gemma-4 family</i>  |                            |                            |                            |                               | $I+ - I-$                     |
| Gemma-4-E2B Base                                                                                           | <b>+2.2</b><br>[+1.9,+2.6] | <b>+2.4</b><br>[+1.8,+3.0] | <b>+2.1</b><br>[+1.6,+2.6] | <b>+0.5</b><br>[+0.4,+0.7]    | <b>+1.5</b><br>[+1.2,+1.7]    |
| Gemma-4-E2B-it                                                                                             | <b>+1.9</b><br>[+1.7,+2.2] | <b>+3.5</b><br>[+3.1,+3.9] | <b>+0.6</b><br>[+0.3,+0.9] | <b>+4.0</b><br>[+3.7,+4.3]    | <b>+8.3</b><br>[+7.9,+8.7]    |
| Gemma-4-E4B Base                                                                                           | <b>+0.8</b><br>[+0.5,+1.2] | <b>+1.1</b><br>[+0.5,+1.7] | <b>+0.6</b><br>[+0.1,+1.1] | <b>+1.2</b><br>[+1.1,+1.4]    | <b>+3.4</b><br>[+3.2,+3.7]    |
| Gemma-4-E4B-it                                                                                             | <b>+4.5</b><br>[+4.2,+4.8] | <b>+7.2</b><br>[+6.8,+7.7] | <b>+2.2</b><br>[+1.9,+2.5] | <b>+8.5</b><br>[+8.0,+8.9]    | <b>+17.1</b><br>[+16.5,+17.6] |
|  <i>Minstral family</i> |                            |                            |                            |                               | $I+ - I-$                     |
| Minstral-3-3B Base                                                                                         | <b>+2.2</b><br>[+1.8,+2.5] | <b>+5.0</b><br>[+4.5,+5.5] | <b>-0.2</b><br>[-0.6,+0.2] | <b>+5.3</b><br>[+5.0,+5.6]    | <b>+12.0</b><br>[+11.5,+12.5] |
| Minstral-3-3B Inst.                                                                                        | <b>+3.1</b><br>[+2.8,+3.3] | <b>+5.8</b><br>[+5.3,+6.3] | <b>+0.7</b><br>[+0.4,+1.1] | <b>+6.6</b><br>[+6.2,+6.9]    | <b>+14.5</b><br>[+14.0,+15.1] |

Table 10: LOCALNEWSQA metadata gains with paired bootstrap 95% confidence intervals. Each completed cell reports the point estimate on top and the confidence interval below. Accuracy columns report percentage-point gains on target-locale questions; exact-pair and margin-switch columns report percentage-point gains on ambiguous target/contrast pairs. MAPLE rows compare  $(T+, I+) - (T-, I-)$ , while non-MAPLE rows compare  $I+ - I-$ . Positive values favor metadata conditioning or metadata-formatted inference; dark bold point estimates have intervals that exclude zero, while lighter entries overlap zero.

| Model                                                                                                      | GeoMLaMA<br>Δ acc.           | GOQA<br>Δ acc.             | MMLU-CS<br>Δ acc.          | NormAD<br>Δ acc.            | BLEnD<br>Δ acc.              | WVB<br>EMD red.               |
|------------------------------------------------------------------------------------------------------------|------------------------------|----------------------------|----------------------------|-----------------------------|------------------------------|-------------------------------|
|  <i>MAPLE family</i>      |                              |                            |                            |                             | (T+, I+) – (T-, I-)          |                               |
| MAPLE 1B Base                                                                                              | <b>+15.3</b><br>[+6.0,+26.0] | <b>+1.3</b><br>[+0.4,+2.1] | +0.1<br>[-2.5,+2.7]        | +1.5<br>[0.0,+3.0]          | +5.9<br>[-1.3,+12.7]         | 0.00<br>[-0.02,+0.02]         |
| MAPLE 1B Chat                                                                                              | <b>+5.3</b><br>[-4.7,+14.7]  | <b>-3.1</b><br>[-3.7,-2.5] | <b>-0.1</b><br>[-0.7,+0.5] | <b>+3.8</b><br>[+1.1,+6.4]  | <b>+0.3</b><br>[-2.8,+3.6]   | 0.00<br>[0.00,0.00]           |
| MAPLE 3B Base                                                                                              | <b>+2.0</b><br>[-6.7,+10.7]  | <b>+8.4</b><br>[+7.3,+9.6] | <b>+3.9</b><br>[+0.5,+7.4] | <b>+5.1</b><br>[+2.8,+7.3]  | <b>+13.5</b><br>[+7.1,+19.3] | 0.00<br>[0.00,+0.01]          |
| MAPLE 3B Chat                                                                                              | <b>-4.0</b><br>[-15.3,+6.7]  | <b>-2.9</b><br>[-4.0,-1.9] | <b>+0.3</b><br>[-0.5,+1.2] | <b>-1.6</b><br>[-3.3,+0.4]  | <b>-1.5</b><br>[-4.6,+1.5]   | <b>-0.03</b><br>[-0.04,-0.01] |
|  <i>LLaMA family</i>      |                              |                            |                            |                             | I+ – I-                      |                               |
| LLaMA-3.2-1B Base                                                                                          | <b>-2.7</b><br>[-5.3,-0.7]   | <b>-0.1</b><br>[-0.6,+0.3] | <b>+2.0</b><br>[+0.6,+3.5] | +1.4<br>[-0.6,+3.3]         | 0.0<br>[-2.3,+2.0]           | +0.02<br>[-0.02,+0.05]        |
| LLaMA-3.2-1B Inst.                                                                                         | <b>+5.3</b><br>[-2.7,+12.7]  | <b>-0.2</b><br>[-0.6,+0.1] | <b>-1.0</b><br>[-2.7,+0.5] | 0.0<br>[0.0,0.0]            | <b>-1.3</b><br>[-3.8,+1.5]   | <b>+0.03</b><br>[+0.02,+0.04] |
| LLaMA-3.2-3B Base                                                                                          | <b>-5.3</b><br>[-14.0,+2.7]  | <b>+0.4</b><br>[+0.1,+0.7] | <b>+0.8</b><br>[+0.5,+2.1] | +1.9<br>[-0.1,+4.0]         | <b>+1.5</b><br>[-0.3,+3.3]   | <b>-0.03</b><br>[-0.04,-0.02] |
| LLaMA-3.2-3B Inst.                                                                                         | <b>-4.7</b><br>[-13.3,+3.3]  | <b>-1.0</b><br>[-1.4,-0.5] | <b>-2.9</b><br>[-4.9,-0.9] | <b>-0.5</b><br>[-1.6,+0.5]  | <b>-9.2</b><br>[-12.7,-5.9]  | <b>-0.10</b><br>[-0.13,-0.07] |
|  <i>Qwen2.5 family</i>    |                              |                            |                            |                             | I+ – I-                      |                               |
| Qwen2.5-1.5B Base                                                                                          | <b>-10.7</b><br>[-16.7,-5.3] | <b>-2.1</b><br>[-2.7,-1.5] | +0.8<br>[-0.9,+2.4]        | <b>+5.0</b><br>[+1.9,+8.1]  | <b>-0.8</b><br>[-3.1,+1.3]   | +0.01<br>[0.00,+0.01]         |
| Qwen2.5-1.5B Inst.                                                                                         | <b>-8.7</b><br>[-14.7,-3.3]  | <b>-0.8</b><br>[-1.2,-0.4] | <b>-0.3</b><br>[-1.8,+1.5] | <b>-2.7</b><br>[-3.9,-1.6]  | <b>-1.5</b><br>[-4.3,+1.0]   | <b>-0.02</b><br>[-0.04,-0.01] |
| Qwen2.5-3B Base                                                                                            | 0.0<br>[-6.0,+5.3]           | <b>-0.4</b><br>[-0.8,-0.1] | <b>-0.3</b><br>[-1.9,+1.5] | <b>+9.6</b><br>[+7.7,+11.2] | <b>-1.8</b><br>[-3.8,+0.5]   | <b>-0.02</b><br>[-0.04,-0.01] |
| Qwen2.5-3B Inst.                                                                                           | <b>-2.7</b><br>[-9.3,+4.0]   | <b>-0.2</b><br>[-0.7,+0.2] | 0.0<br>[-1.6,+1.8]         | <b>-0.1</b><br>[-1.2,+1.0]  | <b>+1.3</b><br>[-1.5,+4.1]   | <b>+0.01</b><br>[0.00,+0.02]  |
|  <i>Qwen3.5 family</i>    |                              |                            |                            |                             | I+ – I-                      |                               |
| Qwen3.5-0.8B Base                                                                                          | <b>-1.3</b><br>[-5.3,+2.7]   | 0.0<br>[-0.4,+0.4]         | +0.8<br>[-0.8,+2.3]        | <b>-0.5</b><br>[-1.5,+0.5]  | <b>-0.8</b><br>[-2.8,+1.3]   | 0.00<br>[-0.01,+0.01]         |
| Qwen3.5-0.8B Chat                                                                                          | <b>-2.7</b><br>[-6.7,+0.7]   | <b>-0.2</b><br>[-0.5,+0.1] | <b>+1.0</b><br>[-0.3,+2.1] | <b>-2.1</b><br>[-3.0,-1.3]  | <b>-2.0</b><br>[-4.3,+0.3]   | <b>+0.04</b><br>[+0.02,+0.06] |
| Qwen3.5-2B Base                                                                                            | <b>-3.3</b><br>[-6.7,0.0]    | <b>+0.2</b><br>[-0.1,+0.6] | <b>+0.6</b><br>[-0.8,+2.1] | <b>-2.5</b><br>[-3.6,-1.4]  | <b>+1.0</b><br>[-1.3,+3.1]   | <b>+0.01</b><br>[0.00,+0.02]  |
| Qwen3.5-2B Chat                                                                                            | <b>+1.3</b><br>[-3.3,+6.0]   | 0.0<br>[-0.4,+0.4]         | <b>+0.8</b><br>[-0.8,+2.4] | <b>-4.0</b><br>[-4.9,-3.2]  | <b>+0.3</b><br>[-2.0,+2.5]   | <b>-0.01</b><br>[-0.03,+0.01] |
| Qwen3.5-4B Base                                                                                            | <b>+2.0</b><br>[-4.0,+8.7]   | <b>-0.2</b><br>[-0.5,0.0]  | <b>+0.5</b><br>[-0.5,+1.4] | <b>+0.4</b><br>[+0.2,+0.7]  | <b>+0.5</b><br>[0.0,+1.3]    | 0.00<br>[0.00,0.00]           |
| Qwen3.5-4B Chat                                                                                            | <b>-3.3</b><br>[-6.0,-0.7]   | <b>-0.2</b><br>[-0.5,+0.1] | <b>-0.4</b><br>[-1.5,+0.6] | 0.0<br>[0.0,0.0]            | <b>-1.5</b><br>[-2.8,-0.5]   | 0.00<br>[0.00,0.00]           |
|  <i>Gemma-4 family</i>  |                              |                            |                            |                             | I+ – I-                      |                               |
| Gemma-4-E2B Base                                                                                           | <b>+2.7</b><br>[-1.3,+6.7]   | <b>+0.3</b><br>[-0.1,+0.6] | +0.3<br>[-1.3,+1.8]        | <b>+6.1</b><br>[+4.7,+7.5]  | <b>-0.3</b><br>[-2.3,+1.5]   | 0.00<br>[0.00,0.00]           |
| Gemma-4-E2B-it                                                                                             | <b>+2.0</b><br>[-5.3,+9.3]   | <b>-0.1</b><br>[-0.4,+0.3] | <b>+0.5</b><br>[-1.4,+2.7] | <b>+1.1</b><br>[+0.1,+2.2]  | <b>-0.5</b><br>[-2.5,+1.5]   | <b>+0.03</b><br>[+0.01,+0.05] |
| Gemma-4-E4B Base                                                                                           | <b>+5.3</b><br>[-2.0,+13.3]  | <b>-0.6</b><br>[-0.9,-0.1] | <b>+2.9</b><br>[+1.4,+4.7] | <b>+1.7</b><br>[+0.9,+2.6]  | <b>+5.1</b><br>[+2.3,+7.9]   | 0.00<br>[0.00,+0.01]          |
| Gemma-4-E4B-it                                                                                             | <b>+1.3</b><br>[-3.3,+6.0]   | <b>-1.0</b><br>[-1.4,-0.6] | <b>+2.7</b><br>[+0.8,+4.5] | <b>+2.5</b><br>[+1.2,+3.8]  | <b>-1.5</b><br>[-4.1,+0.8]   | <b>-0.01</b><br>[-0.02,+0.01] |
|  <i>Minstral family</i> |                              |                            |                            |                             | I+ – I-                      |                               |
| Minstral-3-3B Base                                                                                         | <b>-3.3</b><br>[-8.7,+1.3]   | <b>-0.3</b><br>[-0.7,0.0]  | 0.0<br>[-1.3,+1.4]         | <b>-2.1</b><br>[-3.0,-1.0]  | <b>-0.5</b><br>[-3.1,+2.0]   | <b>-0.04</b><br>[-0.06,-0.01] |
| Minstral-3-3B Inst.                                                                                        | <b>-12.7</b><br>[-18.0,-7.3] | <b>+0.1</b><br>[-0.3,+0.4] | <b>-2.7</b><br>[-4.4,-1.1] | <b>-4.2</b><br>[-5.1,-3.3]  | <b>-0.8</b><br>[-2.5,+0.8]   | <b>-0.02</b><br>[-0.03,0.00]  |

Table 11: External benchmark metadata gains with paired bootstrap 95% confidence intervals, grouped by model family. Each completed cell reports the point estimate on top and the confidence interval below. GEOMLAMA, GLOBALOPINIONQA (GOQA), GLOBAL-MMLU-CS, NORMAD, and BLEND entries are percentage-point accuracy gains. WORLDVALUESBENCH (WVB) reports Earth Mover Distance reduction, computed as metadata-free EMD minus metadata-formatted EMD, so positive values are better in every column. MAPLE rows compare (T+, I+) – (T-, I-); all other rows compare I+ – I-. Dark bold point estimates have intervals that exclude zero, while lighter entries overlap zero.

| Metric                                                                                     | (T+, I+)     | (T-, I-) |
|--------------------------------------------------------------------------------------------|--------------|----------|
| <i>Mean distance to the requested country’s human WVS map coordinate (lower is better)</i> |              |          |
| MAPLE 1B                                                                                   | <b>1.416</b> | 6.914    |
| MAPLE 3B                                                                                   | <b>2.606</b> | 5.794    |

Table 12: Appendix-only country-conditioned World Values Survey projection on the 15 countries overlapping the MAPLE training set. We compare the primary MAPLE settings under a bundled metadata prompt with soft country framing and short labeled WVS answer options. Top-3 retrieval remains flat at 3/15 countries, so we report the more diagnostic target-distance metric.

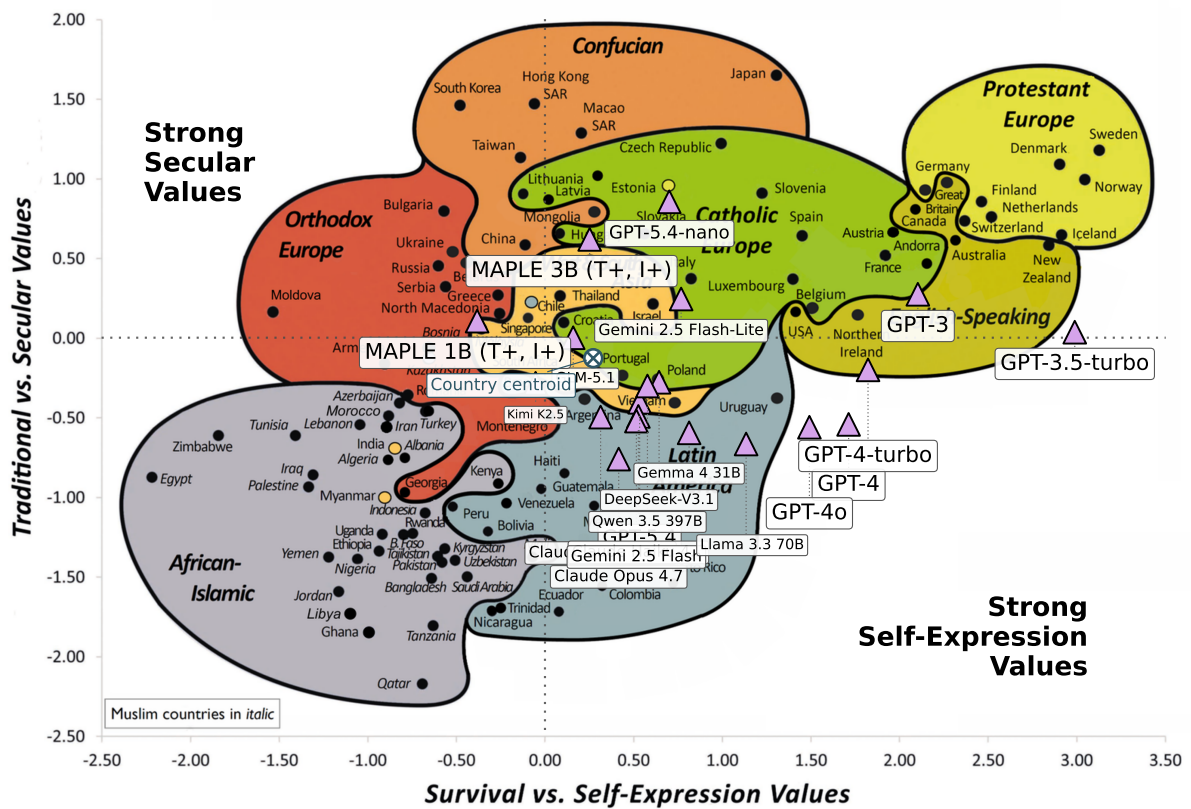


Figure 18: Appendix-only all-country WVS map using the official axes. Released reference-model centroids are shown alongside the MAPLE centroids averaged over the corresponding country-conditioned sweep. The metadata-conditioned MAPLE centroids fall within the main vendor cloud and remain far closer to the human-country centroid than the metadata-free MAPLE controls, which stay off-map.

## A.5 Additional Results

Figures 19-25 collect the remaining supporting views for the main controlled language-modeling studies. They include country-level local performance, asymmetry across continent pairs, the 500M counterpart to the local-vs-global comparison, both leave-one-out controls, and the supplementary 500M cross-continent plot. We place them together here so that each supporting result remains close to the figure that substantiates it.

| Continent | Avg. doc len. | Vocab overlap w/ Africa | Entity overlap w/ Africa | Top theme share |
|-----------|---------------|-------------------------|--------------------------|-----------------|
| Africa    | 243           | 1.000                   | 1.000                    | 0.756           |
| America   | 341           | 0.734                   | 0.149                    | 0.572           |
| Asia      | 254           | 0.770                   | 0.210                    | 0.634           |
| Europe    | 278           | 0.745                   | 0.206                    | 0.569           |

Table 13: Corpus statistics used to interpret the Africa asymmetry in Figure 20. Values come from balanced 20k-document samples per continent. Africa has shorter documents, substantially lower cross-continent named-entity overlap, and much heavier concentration in one dominant theme (*Development & Society*), making transfer out of Africa less symmetric than transfer into Africa.

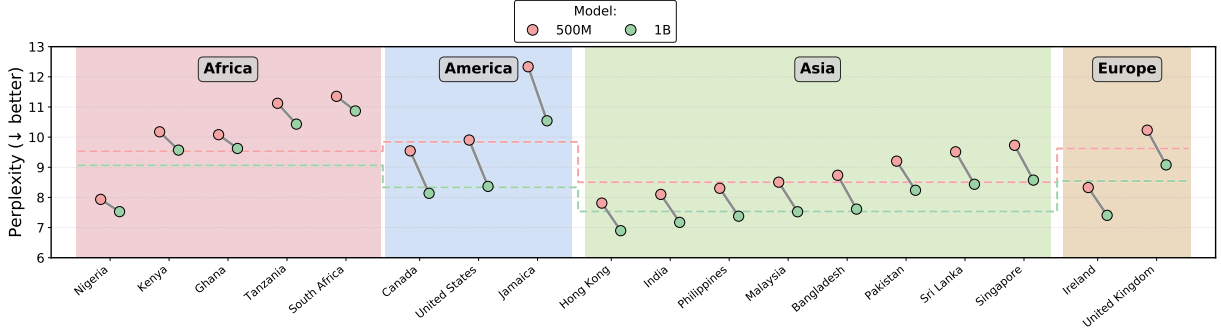


Figure 19: Country-level in-region perplexity for  $\text{LOCAL}(\text{T}^+, \text{I}^+)$ , aggregated within each country by micro-averaged perplexity for the local 500M and 1B models.

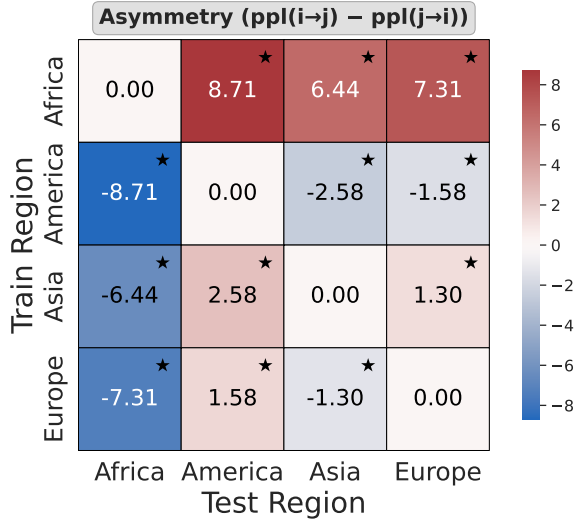


Figure 20: Cross-continent asymmetry for the 1B  $\text{LOCAL}(\text{T}^+, \text{I}^+)$  local models. A higher value indicates that model trained on continent  $i$  when evaluated on continent  $j$  has a higher perplexity than when a model trained on continent  $j$  is evaluated on continent  $i$ .

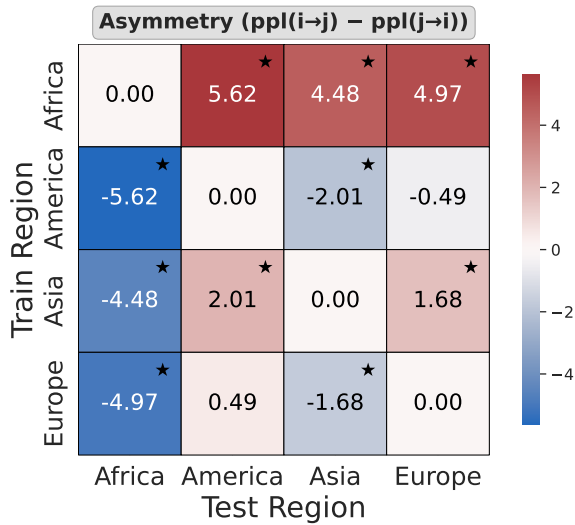


Figure 21: Cross-continent asymmetry for the 500M  $\text{LOCAL}(\text{T}^+, \text{I}^+)$  local models. A higher value indicates that model trained on continent  $i$  when evaluated on continent  $j$  has a higher perplexity than when a model trained on continent  $j$  is evaluated on continent  $i$ .

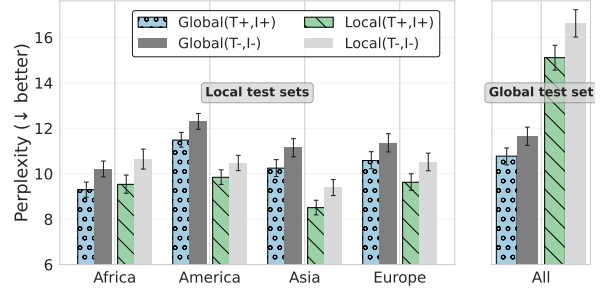


Figure 22: Local-vs-global perplexity for the 500M model family.  $\text{GLOBAL}(\text{T}^+, \text{I}^+)$  models conditioned with metadata during training and inference have lower perplexities than the control variant  $\text{GLOBAL}(\text{T}^-, \text{I}^-)$ , and have similar perplexities to the  $\text{LOCAL}(\text{T}^+, \text{I}^+)$  models on local test sets.

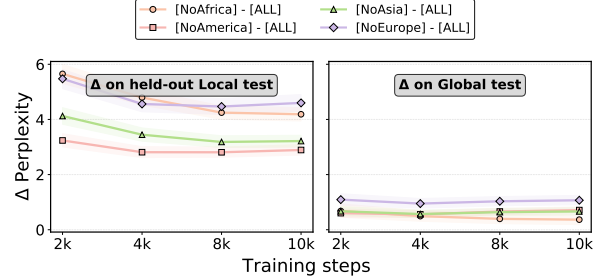


Figure 23: Leave-one-out global models exhibit similar increases in perplexity relative to the full  $\text{GLOBAL}(\text{T}^+)$  model when evaluated on both the held-out continent and the global test set, indicating that training data from all regions is necessary for overall performance. All differences are statistically significant (95% CI).

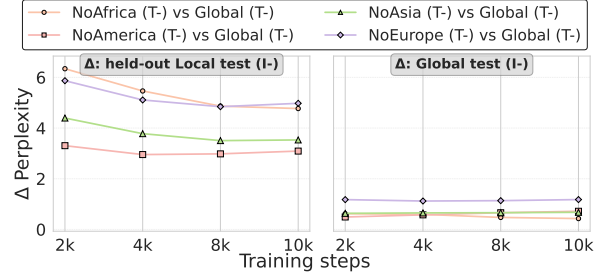


Figure 24: Metadata-free leave-one-out ablations. Leave-one-out global models exhibit similar increases in perplexity relative to the full  $\text{GLOBAL}(\text{T}^+, \text{I}^+)$  model when evaluated on both the held-out continent and the global test set, indicating that training data from all regions is necessary for overall performance.

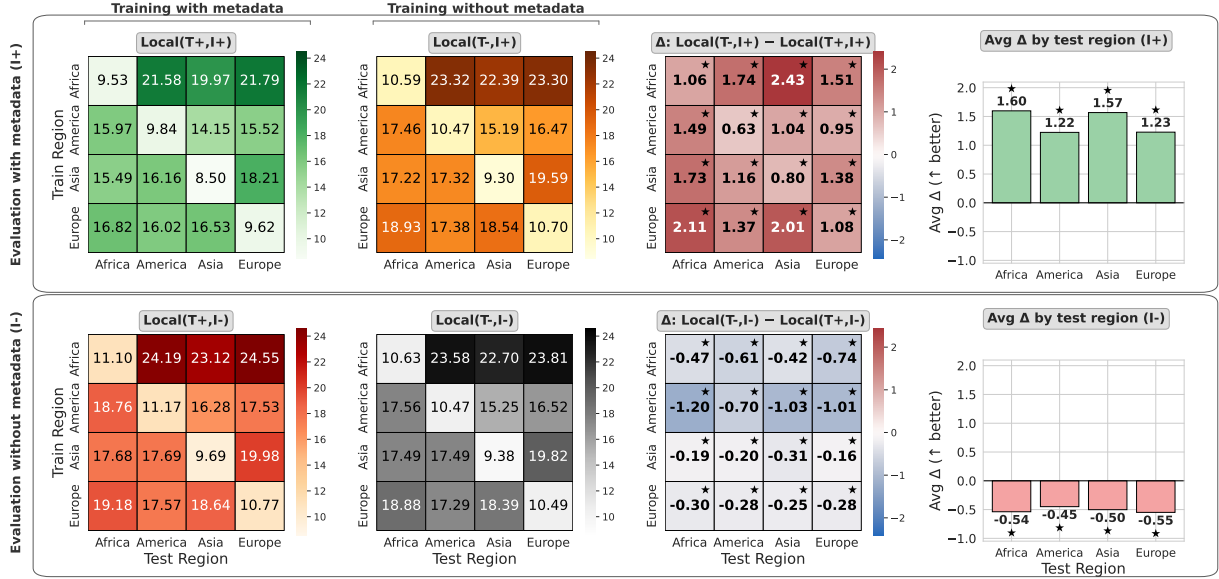


Figure 25: Cross-continent perplexity for the 500M local models. **LOCAL(T+,I+)** models have higher perplexities on cross-continent test sets (off-diagonal) than on local ones. Positive differences between perplexities of models trained without metadata and with metadata on the same test sets indicate the effectiveness of metadata in improving cross-region generalization while maintaining local performance.

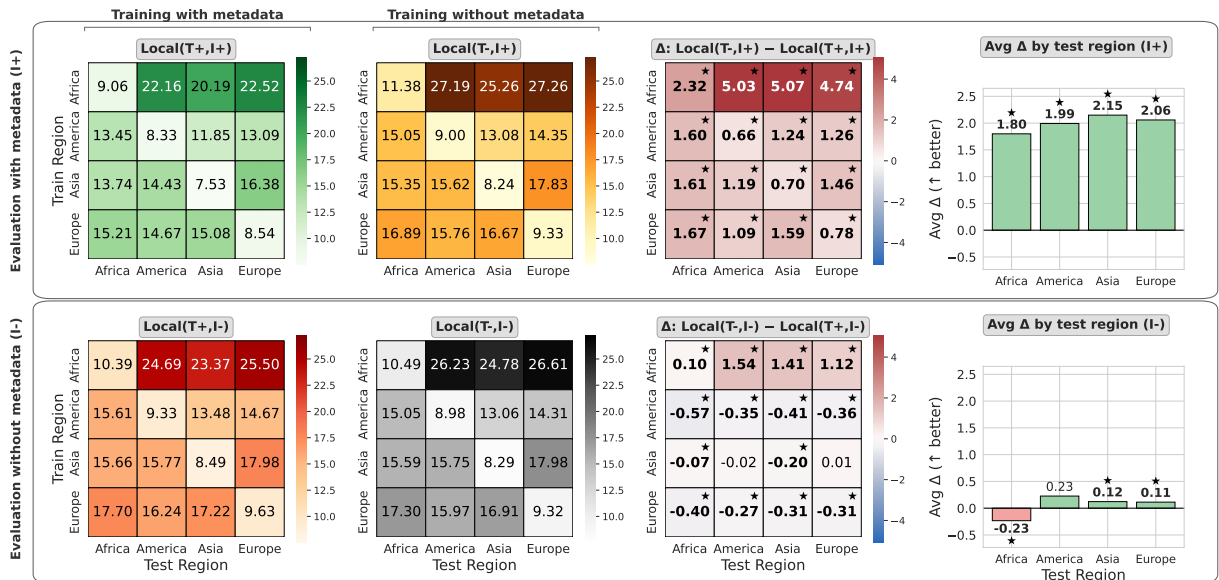


Figure 26: Full cross-continent diagnostic view for the four 1B local model configurations. The first two columns show the raw heatmaps for the I+ and I- settings; the third column shows same-setting deltas, and the fourth shows average delta by test region. Positive  $\Delta$  favors metadata conditioning.

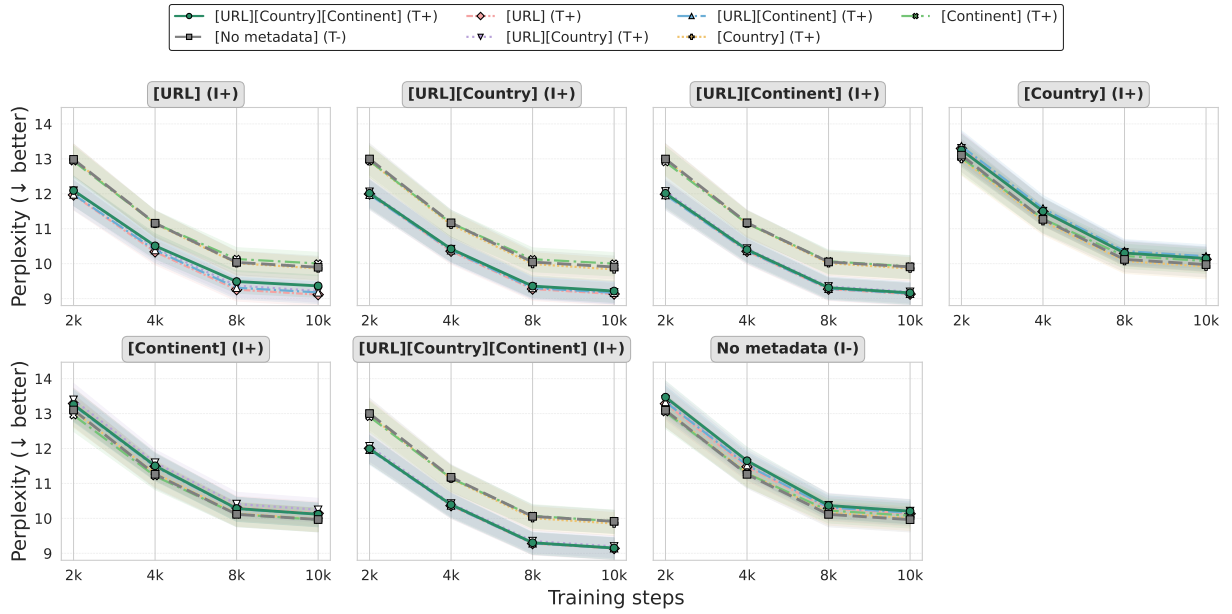


Figure 27: Expanded metadata ablations. Full checkpointed results for [URL], [URL][Country], [URL][Continent], [Country], and [Continent], evaluated on the five metadata-formatted test sets, the [URL][Country][Continent] test set, and the no-metadata test set. The same overall pattern persists across intermediate checkpoints, and [URL] remains the most effective single-metadata variant.

**Scale extension to 3B, compute, and token efficiency.** Figure 28 summarizes the final global-model tradeoff between pretraining compute and downstream performance for the 1B and 3B families. We plot global perplexity together with downstream QA accuracy, using estimated training FLOPs ( $6ND$ ) because the 3B runs used a separate  $4\times B200$  setup while the 500M and 1B families used  $4\times A100$ s. Absolute runtime differs accordingly. The successful 3B pretraining runs used a single node with four B200 GPUs, whereas the smaller families used single nodes with four A100 GPUs. The optimizer, schedule, tokenizer, and data pipeline remain the same, so we interpret this hardware difference as relevant to throughput rather than to the factorial comparison itself. For an apples-to-apples comparison, the 3B QA points use the same SFT protocol as the 1B appendix comparison. Figure 6 complements this view by showing the training-token trajectories for 500M, 1B, and 3B. Across all three scales, `GLOBAL(T+,I+)` reaches the final `GLOBAL(T-,I-)` perplexity using fewer training tokens, showing that metadata improves optimization efficiency rather than only the final value.

**Adversarial metadata corruption during downstream inference.** The main downstream experiment uses correct locale metadata at inference. To test robustness, we repeat the same benchmark while corrupting the URL (and therefore the corresponding country and continent tags) for 25%, 50%, 75%, and 100% of the evaluation samples. Figure 29a reports full-set target accuracy under corruption, and Figure 29b reports the ambiguous split only. Under both views, the metadata-conditioned run declines only modestly as corruption increases, while the metadata-free control remains nearly flat because it does not consume metadata at inference. This pattern is consistent with the

| Condition                | 1B Overall | 1B Ambiguous | 3B Overall | 3B Ambiguous |
|--------------------------|------------|--------------|------------|--------------|
| Clean (T+, I+)           | 34.34      | 38.70        | 35.53      | 37.69        |
| Full URL mismatch (100%) | 34.10      | 38.27        | 34.89      | 36.64        |
| URL-country mask (100%)  | 34.30      | 38.67        | 34.72      | 36.61        |
| No metadata (T-, I-)     | 32.17      | 34.66        | 31.09      | 30.22        |

Table 14: Summary of the adversarial URL study. The URL-country-mask condition removes the country-code portion of the URL while preserving the rest of the prompt, isolating how much of the gain comes from the most literal country cue.

main `LOCALNEWSQA` factorial, where metadata-aware pretraining appears to account for more of the gain than the corrupted inference-time URL cue. Table 14 gives the clean-versus-corrupted comparison. At 1B, full corruption reduces overall accuracy from 34.34% to 34.10%, still above the metadata-free control at 32.17%, while the narrower URL-country-mask variant ends at 34.30%. At 3B, full corruption reaches 34.89% overall and the URL-country-mask variant reaches 34.72%, both still well above the metadata-free control at 31.09%. The same pattern holds on the ambiguous split, where the masked URL variant stays much closer to the clean run than to the metadata-free control. Taken together, these results indicate that country-code cues matter, but broader URL structure also contributes.

**Country-level breakdowns.** The local-model analyses in Section 5.2 operate on continent-level train/test splits, but the `with_metadata` test samples preserve the underlying `COUNTRY`: tag in the serialized text. This lets us aggregate performance by country after evaluation. For perplexity, we micro-average by summing token-level losses and token counts within each country before exponentiating. The main paper’s Figure 19 includes this country-level view for the local 500M and 1B models. Figures 30 and 31 provide analogous country-level downstream accuracy breakdowns for the global chat models after SFT.

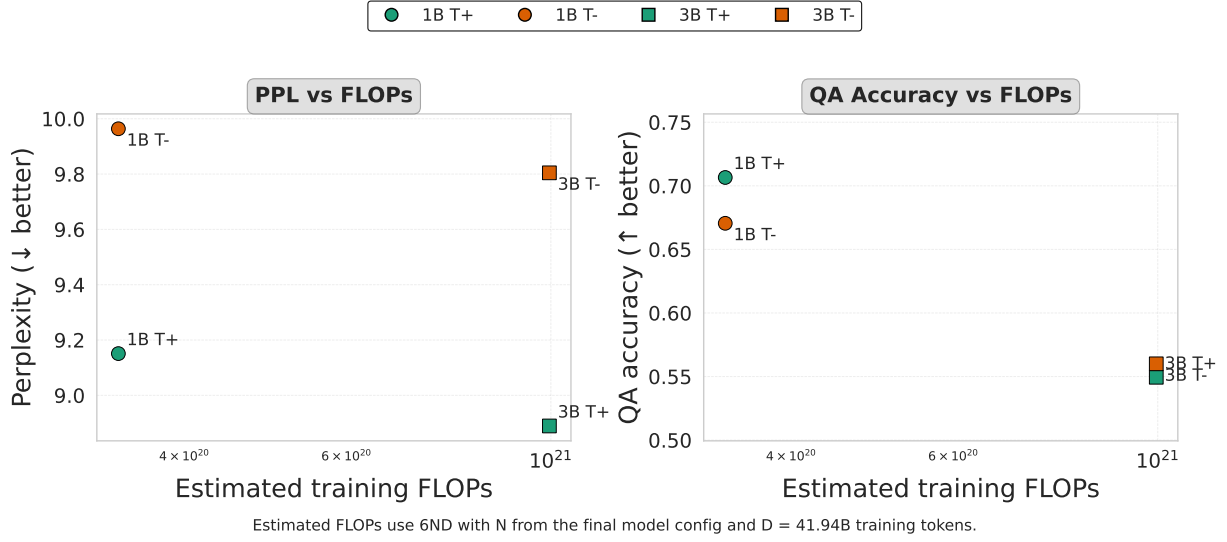


Figure 28: Tradeoff between final performance and estimated training FLOPs for the global 1B and 3B families. The left panel shows global perplexity, and the right panel shows downstream QA accuracy. The 3B QA points use the same SFT protocol as the 1B appendix comparison.

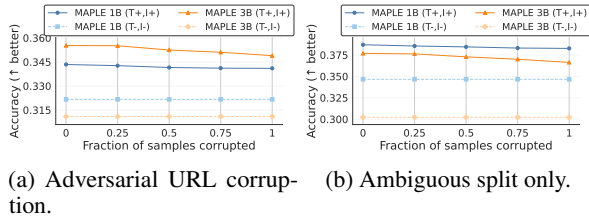


Figure 29: Adversarial downstream evaluation with pretrained closed-form scoring. Increasing metadata corruption causes a monotonic decline for metadata-conditioned inference, while metadata-free inference is largely unchanged.

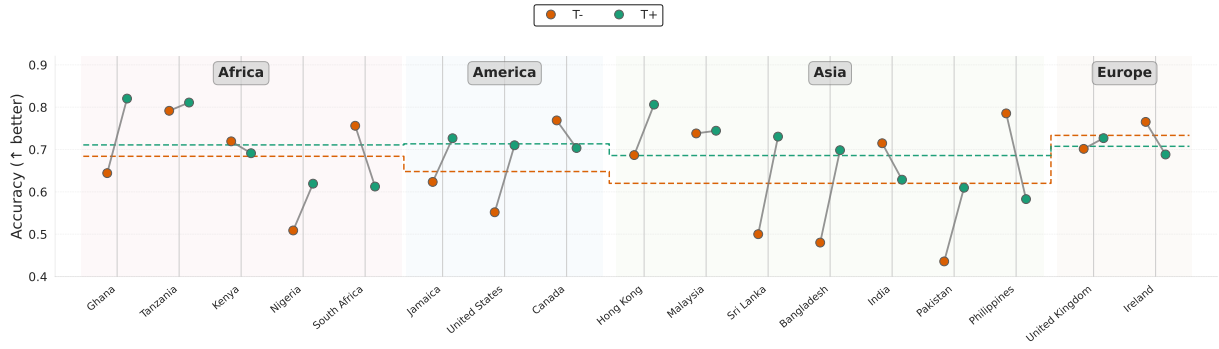


Figure 30: Country-level downstream QA accuracy for the 1B chat models. Within each country, **GLOBAL(T+, I+)** and **GLOBAL(T-, I-)** are compared over the same question pool.

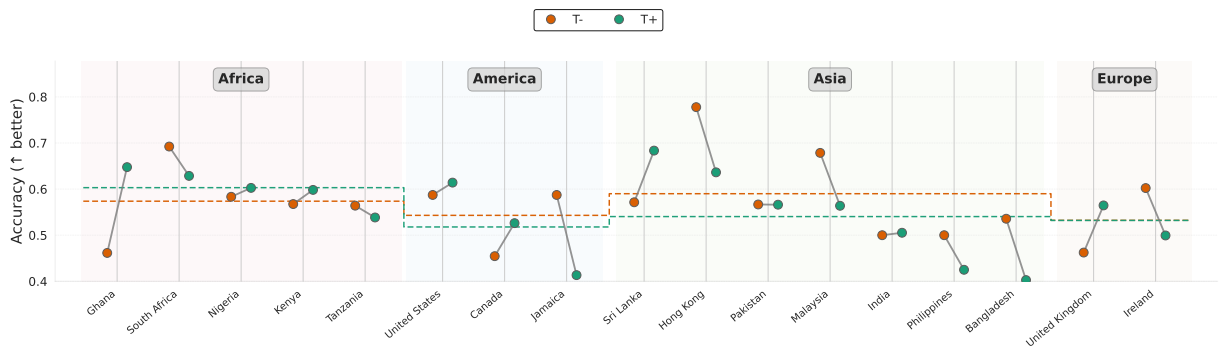


Figure 31: Country-level downstream QA accuracy for the 3B chat models. The larger model family shows stronger country-to-country variance.